

**ENHANCING HOSPITAL NETWORK RESILIENCE
USING REDUNDANCY PROTOCOLS**

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**THIS THESIS IS SUMMITTED IN PARTIAL FULFILLMENT
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DECLARATION

I hereby declare that the report is based on my original work except for quotations and citations, which have been duly acknowledged. I also declare that it has not been previously or concurrently submitted for any other degree at Universiti Sultan Zainal Abidin or other institutions.

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In the Name of Allah, the Most Gracious and the Most Merciful.

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May Allah SWT bless all the efforts that have been made in completing this report.

Thank you.

LIST OF ABBREVIATIONS

HSRP	Hot Standby Router Protocol
RSTP	Rapid Spanning Tree Protocol
PDR	Packet Delivery Ratio
VLAN	Virtual Local Area Network
HQ	Headquarter
OSPF	Open Shortest Path First
DHCP	Dynamic Host Control Protocol
SSH	Secure Shell
PAT	Patients Access Token
ACL	Access Control List
NAT	Network Addressing Translation
EHR	Electronic Health Record
SDLC	System Development Life Cycle
MLS	Multilayer Switches
IP	Internet Protocol
MAC	Media Access Control
BPDU	Bridge Protocol Data Units
LACPDU	Link Aggregation Protocol Data Unit
VRRP	Virtual Router Redundancy Protocol
GLBP	Gateway Load Balancing Protocol
FHRP	First Hop Routing Protocol
STP	Spanning Tree Protocol
MTTF	Mean Time to Failure
MTTR	Mean Time to Repair
bps	Bit per second

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ABSTRACT

Hospital networks require high availability and reliability to support critical healthcare services, where network failures may disrupt clinical systems and compromise patient safety. This project aims to enhance the resilience of an existing hospital network found in a paper called **Smart Hospital Network Enterprise Design for Medicine City Hospital via Packet Tracer** by Mohammed Rajih Jassim (2024) by implementing redundancy protocols in a simulated two-tier (collapsed core) enterprise network environment.

The network was enhanced using Hot Standby Router Protocol (HSRP) for default gateway redundancy and Rapid Spanning Tree Protocol (RSTP) for fast Layer 2 convergence and loop prevention. Network performance was evaluated based on availability, failover time, packet loss and packet delivery ratio (PDR) under controlled failure scenarios. The results show that the proposed redundancy mechanisms improve network availability, reduce failover time and lower packet loss and higher PDR, demonstrating their effectiveness in enhancing hospital network resilience.

ABSTRAK

Rangkaian hospital memerlukan tahap ketersediaan dan kebolehpercayaan yang tinggi bagi menyokong perkhidmatan kesihatan kritikal, di mana kegagalan rangkaian boleh mengganggu sistem klinikal dan menjejaskan keselamatan pesakit. Projek ini bertujuan untuk meningkatkan ketahanan rangkaian hospital sedia ada yang diadaptasi daripada kajian **Smart Hospital Network Enterprise Design for Medicine City Hospital via Packet Tracer** oleh Mohammed Rajih Jassim (2024) melalui pelaksanaan protokol redundansi dalam persekitaran rangkaian perusahaan dua lapisan (collapsed core) yang disimulasikan.

Rangkaian tersebut dipertingkatkan dengan menggunakan Hot Standby Router Protocol (HSRP) bagi menyediakan redundansi gerbang lalai serta Rapid Spanning Tree Protocol (RSTP) bagi memastikan penumpuan pantas dan pencegahan gelung pada Lapisan 2. Prestasi rangkaian dinilai berdasarkan throughput, ketersediaan dan masa peralihan (failover time) di bawah senario kegagalan terkawal. Keputusan menunjukkan bahawa mekanisme redundansi yang dicadangkan berjaya meningkatkan ketersediaan rangkaian, mengurangkan masa peralihan dan pengurangan kehilangan paket dan peningkatan Nisbah penghantaran paket, sekali gus membuktikan keberkesanannya dalam meningkatkan ketahanan rangkaian hospital.

CHAPTER 1

INTRODUCTION

This chapter will provide an overview of the project by introducing the context, rationale and concepts related to a resilient hospital network. Background of a resilient hospital network infrastructures along with the limitations are highlighted.

1.1 Project Background

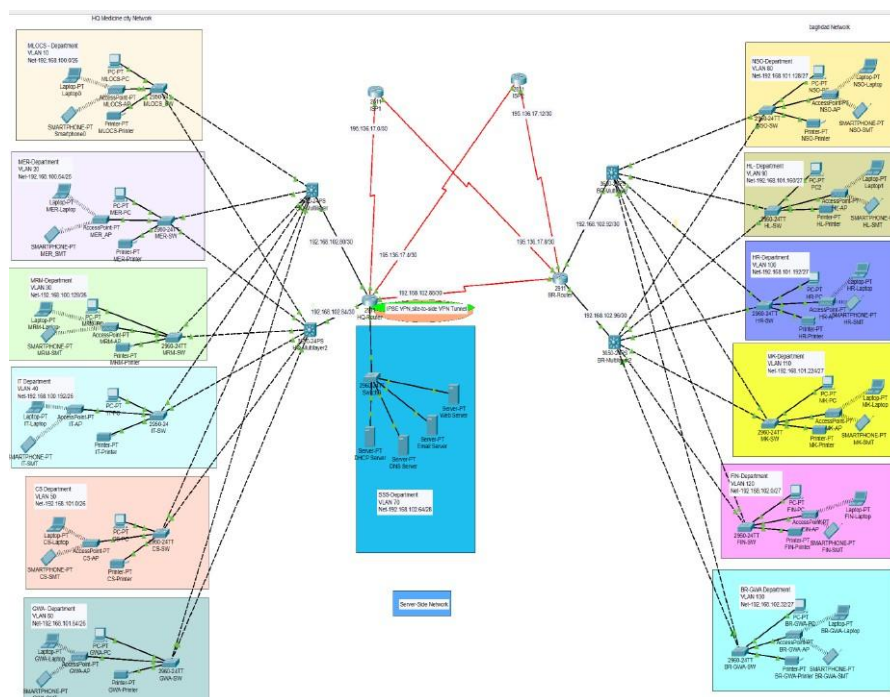
Healthcare networks must ensure high availability and efficient network performance to prevent service disruption that could lead to life-threatening incidents (Rodrigues et al., 2021). Studies show that healthcare organizations experience 4 to 6 network downtime incidents annually, with each lasting 2.5 to 4 hours on average (Proactive Chart, 2025). High availability is measured using the “nines of availability” metric as shown in *Table 1.1* with 99.999% (Five Nines) being the golden standard for industry (Joe Hertvik, 2024).

Availability	Downtime / Year	Downtime / Month	Downtime / Week	Downtime / Day
99.999%	5.256 minutes	0.438 minutes	0.101 minutes	0.014 minutes
99.995%	26.28 minutes	2.19 minutes	0.505 minutes	0.072 minutes
99.990%	52.56 minutes	4.39 minutes	1.011 minutes	0.144 minutes
99.950%	4.38 hours	21.9 minutes	5.054 minutes	0.72 minutes
99.900%	8.76 hours	43.8 minutes	10.108 minutes	1.44 minutes
99.500%	43.8 hours	3.65 hours	50.538 minutes	7.2 minutes
99.250%	65.7 hours	5.475 hours	75.808 minutes	10.8 minutes
99.000%	87.6 hours	7.3 hours	101.077 minutes	14.4 minutes

Table 1.1:"Nines of Availability" table

There are several key practices that could minimize network downtime and improve network resilience. The first one is **network segmentation** which refers to the isolation of critical systems from other systems as to minimize impacts from failures. **Redundant path**, on the other hand, eliminates single point of failure by deploying multiple devices and links (IT Strategy News, 2025).

This project is proposed to enhance the resilience of an existing hospital network design found in a paper named **Smart Hospital Network Enterprise Design for Medicine City Hospital via Packet Tracer** by Mohammed Rajih Jassim (2024). The network design for the smart hospital from the paper is as shown in *Figure 1.1*. It is later improved by implementing redundancy protocols like **Hot Standby Router Protocol (HSRP)** and **Rapid Spanning Tree Protocol (RSTP)**. The results before and after redundancy protocols implementation will then be compared.



1.3 Objectives

The objectives of this project are as follows:

1. To **study** the role of redundant protocols techniques in improving network resilience and availability within a hospital environment.
2. To **develop** a resilient hospital network design using HSRP for gateway redundancy mechanisms and RSTP for quicker failover implemented in Cisco Packet Tracer.
3. To **compare** the performance of the enhanced network against the existing network design based on key performance metrics, including network availability, failover time, packet loss and packet delivery ratio (PDR).

1.4 Scope

In this project, the network is designed and simulated using **Cisco Packet Tracer**. The network is using **2-tier network architecture** consisting of distribution and access layer. The hospital network consists of **13 Virtual Local Area Networks (VLANs)**. 6 (VLAN 10-60) is on Headquarters (HQ) side, 6 (80-130) on Branch side and 1 VLAN (VLAN 70) for containing all servers. In the original network, protocols like Open Shortest Path First (OSPF) and Dynamic Host Configuration Protocol (DHCP), along with security features like Secure Shell (SSH), Patients Access Token (PAT) combined with Access Control Lists (ACL), Network Addressing Translation (NAT) and IPSec Site-to-Site Virtual Private Network (VPN) were already configured.

An additional of 2 redundancy protocols are configured in this project to improve network resilience, thus improving its availability. **HSRP** is configured to provide **gateway redundancy**. **RSTP** and a **redundant link linking MLS 1 and MLS 2** are implemented to enhance **link redundancy and ensure fast convergence**.

Since the **project is fully a simulation**, the results might not be able to fully showcase an accurate real-world scenario where many factors may affect the network. Thus, this research only acts as a reference for real-world deployment. Network performance is evaluated based on **availability, failover time, packet loss and PDR**.

This research is confined to evaluating the performance of First-Hop Redundancy (HSRP) and Layer 2 convergence (RSTP) within a two-tier hospital network architecture. Because these protocols operate entirely within the local area network (LAN), the testing boundary is strictly limited to the access and distribution layers. Consequently, inter-site traffic, edge security protocols and Wide Area Network (WAN) routing mechanisms such as OSPF fall outside the scope of this evaluation.

Furthermore, as the Headquarters (HQ) and Branch networks feature identical structural configurations, performance simulations and data collection are conducted exclusively on the HQ infrastructure to maintain an objective and non-redundant comparative analysis.

1.5 Limitations of Work

There are several limitations encountered when doing this project. Those limitations include simulation constraints and limited security implementation which is further discussed below.

1.5.1 Simulation Constraints

The project is implemented using Cisco Packet Tracer which does not fully reflect real-world network behavior since it is merely a network simulator. It comes with many functional and protocol limitations. For example, some redundancy protocols like Gateway Load Balancing Protocol are not supported in this simulator. Performance

results are based on simulated traffic and non-real hospital workloads.

1.5.2 Limited Security Implementation

This project focuses on network redundancy. Advanced security solutions like Firewall configuration are not featured since the project emphasized more on availability and redundancy rather than full enterprise security.

1.6 Expected Result

After redundancy protocols are implemented, the enhanced network is expected to **achieve higher availability** with minimal service disruption during device or link failures. **Packet loss is expected to decrease**, and **PDR is expected to be higher** when link or device failure occurs. **Network recovery or failover time is expected to occur within a few seconds** due to HSRP, RSTP and a redundant path connecting MLS 1 and 2.

1.7 Gantt Chart

The project timeline is as presented in *Figure 1.2* below.

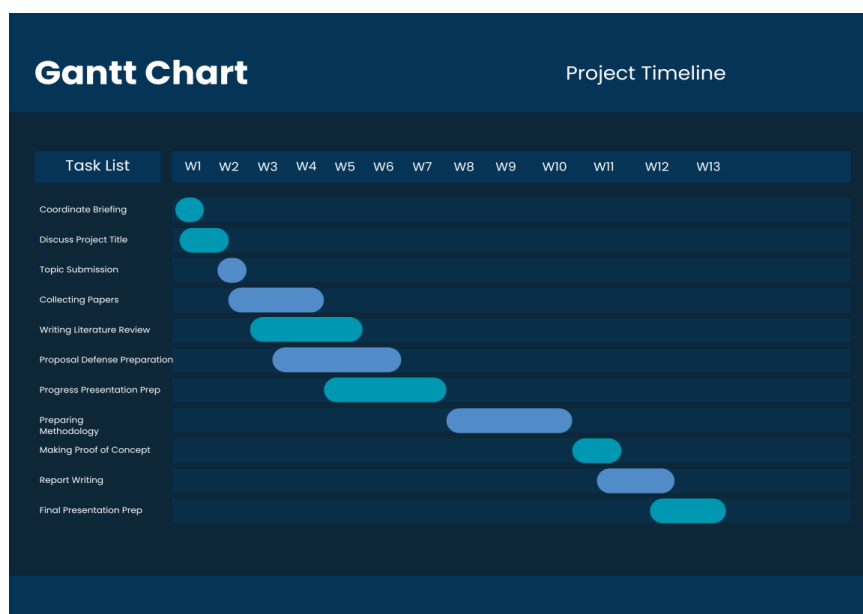


Figure 1.2: Project Gantt Chart

1.8 Summary of the Chapter

This chapter provided the background and motivation for enhancing hospital network resilience. The importance of having high availability in healthcare networks is highlighted and the lack of redundancy in the existing network design is identified as a key problem. Apart from that, this chapter also outlined the project objectives, scope, limitations and expected outcomes, emphasizing the importance of implementing redundancy protocols to improve network availability and reliability in healthcare network.

CHAPTER 2

LITERATURE REVIEW

2.1 Introduction

This chapter covers the hospital network architecture and resilience, redundancy protocols and mechanisms used and details on the performance metrics.

2.2 Hospital Network Infrastructure and Resilience

Hospitals today rely on multiple systems like EHR, lab interfaces and even communication platforms for daily operation. If a network is interrupted or goes down, these systems would become inaccessible or put to a halt (DB Technology, 2023; Sameer Malik, 2025). As a result, patient care would be compromised, treatments get delayed and the waiting line will be excessively long (DB Technology, 2023).

Network redundancy is one of many solutions implemented by several organizations to mitigate issues related to availability (Octavian, 2024). Rodrigues (2021) stated that high availability is crucial to guarantee quality of services in healthcare system. In the same paper, it is mentioned that the availability model of Stochastic Petri Net shows the highest availability in a scenario where redundancy is implemented, reaching up to 99.9199% or 7.01 downtime annually (Rodrigues et al., 2021).

2.3 Network Redundancy Mechanisms

Network redundancy provides alternative routes or devices to take over the task in case the active ones go down or compromised to reach high availability (Octavian, 2024).

Availability and redundancy must be achieved to prevent loss or damage to critical data like patients' data stored in EHR. One way to have redundancy in the network is by deploying multiple devices or links in the required locations (Simanjuntak et al., 2023). On the other hand, default gateway redundancy can be achieved by deploying FHRP protocols (Shahriar & Fan, 2020).

2.4 Redundancy Protocols at Different Network Layers

In this project, 3 protocols are chosen to provide redundancy to the hospital network as to make it more resilient and achieve higher availability. The protocols are HSRP and RSTP.

2.4.1 First Hop Redundancy Protocols (FHRP)

FHRP provides backup or alternatives for default gateway, thus preventing network downtime when the primary gateway fails and lets the standby gateway take over seamlessly (Shahriar & Fan, 2020). Shahriar and Fan (2020) have conducted research on the three protocols in FHRP family, Hot Standby Router Protocol, Virtual Router Redundancy Protocol (VRRP) and Gateway Load Balancing Protocol (GLBP). The results are as shown below:

First Hop Redundancy Protocol (FHRP)	Average Consumed Time For no Link Failure (sec)	Average Consumed Time For a Failed Link (sec)
HSRP	57.3	86.6
VRRP	60.1	90
GLBP	61.5	91.5

Table 2.1: Average Time Consumption

From the results, it is noted that HSRP consumed less time on average compared to the other FHRP protocols as shown in *Table 2.1*. However, it is also mentioned that HSRP is more unstable compared to VRRP and GLBP.

TABLE II. AVERAGE PACKET TRANSMISSION RATE

First Redundancy Protocol (FHRP)	Hop	Average Packet/1sec For no Link Failure	Average Packet/1sec For a Failed Link
HSRP		44.25	28.75
VRRP		43.7	24.35
GLBP		42.15	23.96

Table 2.2: Average Packet Transmission Rate

Average Packet Transmission Rate for HSRP is the highest for both normal and link failure operations as shown in *Table 2.2*.



Figure 6. End to end delay performance analysis of FHRP when there is no link failure.

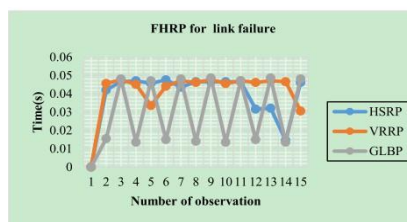


Figure 7. End to end delay performance analysis of FHRP for link failure.

Figure 2.1: End to end delay

Average End to end delay for HSRP is lower than VRRP and GLBP as shown in *Figure 2.1*. The paper then concluded that for small or medium sized companies with complex VLANs, HSRP is recommended due to the overall performance shown.

In another study by Firdaus Syah and Adriyanto (2025) that compare HSRP with VRRP found that HSRP failover speed to be quicker than VRRP's, while VRRP is more flexible since it is open-standard (not tied to Cisco) and more stable.

HSRP is a Cisco Proprietary protocol that provides gateway redundancy by making multiple routers or MLS act like one virtual router, allowing alternate or standby devices to take over the active role automatically in case of current active device failure without the end users noticing it (Octavian, 2024; Rachman Harahap & Handoko, 2021; Yogasetya Suhandi et al., 2022).

From these previous studies, HSRP is chosen out of the other FHRPs as it shows the best results for average time consumption, packet transmission rate, end to end delay and failover speed.

2.4.2 Layer 2 Redundancy using Rapid Spanning Tree Protocol (RSTP)

When creating a more reliable and redundant network, chance of creating a loop increases. Broadcast storms could happen in Layer 2 devices like switches causing the link to be overloaded, thus causing the entire network to go down. To prevent this from occurring, protocols like STP or RSTP are deployed as they provide both redundancy and fault tolerance to the network (Rathod, 2022).

The traditional STP is known to have longer convergence time (time taken to converge failed link to back up link) while RSTP do it almost instantly (Rathod, 2022). Below are comparisons made by several studies on these two protocols:

Fitur	STP (802.1D)	RSTP (802.1w)
Nama Lengkap	Spanning Tree Protocol	Rapid Spanning Tree Protocol
Protokol di output	Protocol ieee	Protocol rstp
Kecepatan konvergensi	Lambat (hingga 30 detik)	Cepat,dalam beberapa detik
Transisi port	Listening-Learning-Forwarding	Langsung ke Forwarding

Table 2.3: STP and RSTP comparison table

From a table found in a paper by Anggi Silalahi et al., (2025), *Table 2.3*, it is observed that traditional STP took around 30 seconds to converge while RSTP convergence time is very fast as it only took a few seconds. It is also mentioned in the table that compared to STP which must go through 3 port transitions due to its careful nature, RSTP goes straight to Forwarding state. Convergence time is proportional to failover time.

Parameters	STP	RSTP
Standard	IEEE 802.1w	IEEE 802.1D
Number of Packet Sent	23	16
Number of Packet Received	11	15
Number of Packet Lost	12	1
Loss	53%	7%

Table 2.4: STP and RSTP Packet Loss Comparison Table

In another study by Rathod 2022), a huge difference can be observed in terms of packet loss between STP and RSTP as shown in *Table 2.4*. RSTP only lost about 7% of the packets while STP lost 53% of the packets. It is also stated that RSTP and STP provided redundancy by providing back up path for packets in case of a link failure by unblocking the blocked link (previously blocked to prevent loops), which goes from the source and leads to destination.

2.5 Performance Metrics for Network Resilience Evaluation

There are 4 performance metrics chosen to measure the network resilience before and after the protocol's implementation.

2.5.1 Availability

According to Samuel Aras Michael & Boy Yuliadi (2025); Shahriar & Fan, (2020), uninterrupted service is an important aspect of modern networking regardless of which institution or corporation the network is. To achieve uninterrupted service or high availability, network must have high redundancy and fault tolerance (Rathod, 2022). In a study by Kurniawan Harun Rasyid et al. (2024) stated that HSRP successfully maintained connection during router failure with zero packet loss in most scenarios.

A study by Rodrigues et al., (2021) mentioned that servers with redundancy in Scenario A achieved the highest availability of 99.9199% (approximately 7 hours downtime/year) and uses the formula:

$$\text{Availability} = \text{MTTF} / (\text{MTTF} + \text{MTTR})$$

where MTTF (Mean Time to Failure) represents the average uptime and MTTR (Mean Time to Repair) represents the average downtime to calculate Availability.

2.5.2 Failover Time

Failover time is defined as the duration required for a redundancy protocol to detect failure and switch traffic from a primary link to a backup path (Samuel Aras Michael & Boy Yuliadi, 2025). It represents the specific time consumption observed during the transition state when a network recovers from a physical link or device failure (Shahriar & Fan, 2020). In optimized designs like "Zero Link Failure," this time is

measured to be under 5 seconds to ensure continuity (Samuel Aras Michael & Boy Yuliadi, 2025).

Failover time describes the speed of the recovery process, whereas downtime refers to the period during which the service is completely unavailable to the user. A minimized failover effectively eliminates downtime. For instance, if the failover occurs rapidly enough (like zero packet loss), the user perceives no disconnection or downtime at all (Kurniawan Harun Rasyid et al., 2024). While downtime is a state of service loss, failover time is a performance metric comparing how quickly protocols like HSRP or STP can restore the active state (Shahriar & Fan, 2020).

Short failover time is critical because modern environments need services to always be readily available where any latency or disconnection can disrupt real-time data transmission (Shahriar & Fan, 2020). High availability and rapid recovery are also described as primary needs to prevent operational interruptions in critical infrastructure by Samuel Aras Michael & Boy Yuliadi (2025).

2.5.3 Packet Loss

Packet loss happens when data packets that travel across a computer network fail to reach their intended destination. This occurrence usually take place during network convergence or when a link failure happens before a backup path becomes fully active (Rathod, 2022). It serves as a primary metric for evaluating the efficiency of redundancy protocols, as protocols with faster responses drop fewer packets during transition states (Rathod, 2022).

Most papers in the literature review calculate packet loss either by counting the exact number of dropped ping packets during a failure event or by expressing it as a percentage of lost packets relative to the total number of packets sent (Rathod, 2022). In standardized Quality of Service (QoS) assessments, packet loss is evaluated against frameworks like the THIPON standard, where a 0% packet loss is strictly required to achieve a "Very Good" (Sangat Bagus) performance rating (Kurniawan Harun Rasyid et al., 2024).

While the provided papers do not specifically examine hospital networks, they heavily emphasize the importance of mitigating packet loss in equivalent real-time, critical infrastructures such as banking and enterprise environments (Simanjuntak et al., 2023). In these critical settings, high packet loss is entirely unacceptable because it leads to data corruption, transaction failures and severe operational disruptions (Rathod, 2022). Minimizing packet loss to near-zero ensures that hardware failures remain invisible to the end-user, protecting the continuous transmission of sensitive, real-time data (Simanjuntak et al., 2023).

2.5.4 Packet Delivery Ratio (PDR)

"Packet Delivery Ratio" is not explicitly defined in the provided literature. However, the papers did extensively discuss its direct equivalents: "Packet Transfer Rate" and "Throughput." In the context of the provided studies, this concept refers to the network's capability to successfully deliver data packets to their destination, reflecting the overall efficiency and throughput stability of the network under both normal and failure conditions (Shahriar & Fan, 2020).

Rather than a traditional PDR formula (Successfully Received / Total Sent), the provided studies measure delivery success practically via Throughput (measured in kbps or Mbps) and Packet Transfer Rate (measured in packets per second) (Kurniawan Harun Rasyid et al., 2024; Shahriar & Fan, 2020). For example, researchers measure the total time consumption required to successfully deliver a set volume of traffic, such as 1000 ICMP ping requests, to determine the network's delivery efficiency (Shahriar & Fan, 2020).

Similar to packet loss, the explicit context of hospitals is absent, but the rule applies to any institution requiring "uninterrupted and irreversible service" (Shahriar & Fan, 2020). Maintaining a high rate of successful packet delivery is critical because modern critical institutions rely on the rapid, large-scale transfer of data without latency or disconnection (Alfian Armawan Sandi et al., 2025). A high packet delivery capacity ensures that even during a failover event, the network can handle the rerouted traffic without dropping the continuous stream of critical operational data (Simanjuntak et al., 2023).

2.6 Literature Review Tables

2.6.1 General Table

NO.	Paper (Author, Year)	Aim/Objectives	Tools/Method	Advantages	Disadvantages
1	Performance Analysis of FHRP in a VLAN Network with STP (Shahriar & Fan, 2020).	To achieve maximum network redundancy and effective failure recovery by analyzing STP and FHRP in a complex multi-VLAN environment	GNS3 software, Wireshark	The research highlights HSRP's superior speed and packet transfer rates and compare it with other FHRP.	The failure scenarios appear limited to link failures, no analysis of total node/device failures or stateful recovery.
2	Network Disaster Recovery Design Using Hot Standby Router Protocol (HSRP) (Rachman Harahap & Handoko, 2021).	To implement a Disaster Recovery Plan (DRP) using HSRP to ensure network availability during failures	GNS3, Cisco Packet Tracer	Proves HSRP can determine the best path during a main line failure	Simulation-based: The results are from a simulator, not physical hardware.
3	Analisa Performa VLAN Dengan Link Aggregation Control Protocol (LACP) Layer 3 Pada Kasus Penentuan Topologi (Alfian Armawan Sandi et al., 2025).	To improve network performance and reliability by implementing VLAN and LACP	GNS3, MikroTik RouterOS	Demonstrates that LACP/Bonding significantly reduces file transfer times (e.g., saving over 3 minutes for a 16GB file)	High Resource Consumption, CPU power; tests showed the router CPU hitting 100%

4	Implementation Analysis of Network Core Redundancy Using HSRP and VRRP (Firdaus Syah & Adriyanto, 2025).	To evaluate how redundancy protocols affect core enterprise network reliability	Cisco Tracer simulations comparing baseline redundancy configs	Packet Tracer vs	Quantifies of failover times and availability improvements	Results may not reflect hardware behavior outside simulated environment
5	Design and Simulation of a Secured Enterprise Network using Cisco Packet Tracer: A case study of the Faculty of Engineering, University of Lagos (Muhammed Babayode, 2024).	To design and simulate a secured enterprise network for the Faculty of Engineering	Cisco Tracer	Packet Tracer	Incorporates EtherChannel to ensure link redundancy and higher bandwidth, addressing failures found in other studies.	The study is limited to a virtual environment (Packet Tracer) and lacks physical hardware implementation
6	Perancangan Jaringan Zero Link Failure pada Layer 2 (Samuel Aras Michael & Boy Yuliadi, 2025).	To design and simulate a Layer 2 network using STP and EtherChannel	Cisco Tracer, Wireshark (Simulation media) , PPIDOO	Packet GNS3	Rapid Recovery: achieved a failover convergence time of < 5 seconds (compared to >10 seconds manually). Throughput increased significantly (from ~2.4 Kbps to 18 Kbps)	Scope Limitation: The research focuses strictly on Layer 2; it does not address Layer 3 redundancy (Routing) or QoS for real-time applications

7	Perancangan Jaringan Redundancy Menggunakan Konsep Etherchannel dan HSRP dengan Intervlan Routing pada PLN UID Jakarta Raya. (Octavian, 2024).	To design a redundant network for PLN UID Jakarta Raya using EtherChannel, HSRP, and InterVLAN Routing	Cisco Tracer	Packet	with 0% packet loss during tests Integrated Solution: successfully combines HSRP (Gateway Redundancy) with EtherChannel (Link Aggregation) to eliminate single points of failure	Specific Scope: The topology is tailored specifically to PLN's needs, which may limit generalizability without modification
8	Implementasi Konfigurasi Spanning-Tree Protocol (STP) dan Rappid Spanning-Tree Protocol (RSTP) di Cisco Packet Tracer (Anggi Silalahi et al., 2025).	To understand and implement STP and RSTP to prevent loops in complex switched networks.	Cisco Tracer.	Packet	Clear Comparison: effectively demonstrates that while both protocols prevent loops, RSTP offers significantly faster convergence (seconds) compared to STP (up to 30 seconds).	Simulation Constraint: The results are bound to the idealized environment of Packet Tracer and may differ slightly from physical hardware behavior regarding exact convergence timing.
9	Perancangan Infrastruktur Jaringan Berbasis Aplikasi Packet Tracer dengan Metode	To design and evaluate a high availability network using HSRP to eliminate single path failures	Cisco Tracer	Packet simulation with redundant	Shows seamless failover with minimal packet loss in simulation	Focuses only on HSRP; doesn't evaluate alternative

	HSRP (Yogasetya Suhandu et al., 2022).		router paths and HSRP protocol		redundancy mechanisms
10	Comparative Analysis of Spanning Tree Protocol and Rapid Spanning Tree Protocol using Packet Tracer (Rathod, 2022).	To compare STP and RSTP to determine which provides better redundancy and fault tolerance	Cisco Packet Tracer Simulation: Created a ring topology with 5 switches and configured trunking.	Quantifiable Results: Provides a clear statistical comparison showing STP had 53% packet loss while RSTP had only 7%	Simulation Based: The results are derived solely from a simulator (Packet Tracer) and may not fully reflect real-world hardware behavior
11	Simulation and Performance Analysis of Network Backup Systems (HSRP) (Simanjuntak et al., 2023).	To assess performance improvements of backup systems using HSRP	Cisco Packet Tracer used for HSRP simulation & performance checks. GNS3 software, Wireshark	Confirms improved uptime and reduced failover delays	Limited analysis depth on network load or advanced redundancy features
12	Performance and Availability Evaluation of an Smart Hospital Architecture (Rodrigues et al., 2021)	To assess the performance and availability of intelligent hospital systems using analytical models, avoiding the high costs and risks of testing on physical equipment.	Stochastic Petri Nets (SPN), Sensitivity Analysis. Simulate performance and availability,	Cost-Effective Planning: Allows administrators to evaluate and optimize complex infrastructure scenarios without purchasing expensive hardware	Validation Limits: The model was validated using synthetic data and low-capacity machines, which may not fully represent the scale or complexity of

13	Simulasi dan Perancangan Redudansi Jaringan Berbasis Cisco Packet Tracer (Rasyid et al., 2024)	To design and simulate HSRP at SMKS Handayani Makassar to ensure network stability and minimize downtime, validating performance against THIPON standards.	Cisco Packet Tracer, Simulated three scenarios (Normal, Failover, Recovery)	High Availability: HSRP successfully maintained connection with zero packet loss during failover scenarios	a real high-end hospital data center. Performance variance: The "Recovery" phase showed slightly lower throughput (2315 kbps) compared to normal operation (2868 kbps).
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Table 2.5: General Literature Review Table

2.6.2 Simulation Table

NO.	Paper (Author, Year)	Simulation Tool(s)	Network Scenario	Key Parameter	Performance Metrics	Key Findings
1	Performance Analysis of FHRP in a VLAN Network with STP (Shahriar & Fan, 2020).	GNS3	Complex network using STP and multiple VLANs (VLAN10, VLAN20) to compare HSRP, VRRP, and GLBP performance	Protocol Efficiency: Time consumption and Packet Transfer Rate under normal vs. link failure conditions	Time Consumption: Total time for 1000 pings (seconds). Transfer Rate: Packets per second	HSRP Superiority: HSRP consistently consumed the least time (57.3s normal, 86.6s failure) and achieved the highest packet transfer rate compared to VRRP and GLBP.
2	Network Disaster Recovery Design Using Hot Standby Router Protocol (HSRP) (Rachman Harahap & Handoko, 2021).	GNS3, Cisco Packet Tracer	Disaster recovery network connecting a Network Operation Center to an Operation Maintenance division, featuring 4 routers, 2 switches, and 8 clients/PCs to simulate a main and backup site	Failover Capability: HSRP state transition (Active/Standby) and IP SLA for link tracking	Availability: Successful failover (Active to Standby status change)	Disaster Mitigation: The implementation of HSRP successfully provided a backup mechanism where a standby router automatically took over traffic when the primary router failed, ensuring continuity

3	Analisa Performa VLAN Dengan Link Aggregation Control Protocol (LACP) Layer 3 Pada Kasus Penentuan Topologi (Alfian Armawan Sandi et al., 2025).	GNS3	Comparison of two network topologies (A & B) connecting a MikroTik Router to a NAS Server, designed to test the efficiency of VLANs combined with LACP (Bonding) for large data transfers.	Data Transfer Load: File sizes of 1GB, 8GB, and 16GB.	Transfer Duration: Time taken to send files (minutes/seconds). Throughput: Data transmission rate.	Stability: The bonding interface provided better stability and lower jitter compared to single-link connections
4	Implementation Analysis of Network Core Redundancy Using HSRP and VRRP (Firdaus Syah & Adriyanto, 2025).	Cisco Packet Tracer	Enterprise core network comparing a baseline single-router design against redundant HSRP and VRRP configurations	Failover time and Service Availability (Uptime)	Failover Time: measured in seconds (avg < 5s). Stability: Packet loss, latency, and jitter.	HSRP demonstrated faster failover speeds, while VRRP offered better cross-vendor compatibility
5	Design and Simulation of a Secured Enterprise Network using Cisco Packet Tracer: A case study of the Faculty of Engineering,	Cisco Packet Tracer	Enterprise network for the Faculty of Engineering at the University of Lagos (UNILAG), utilizing VLANs to segment departments and	Network Security & Addressing: Implementation of Access Control Lists (ACLs) for	Connectivity Verification: Successful ping replies between authorized VLANs.	Operational Efficiency: VLAN segmentation reduced broadcast traffic and improved network

	University of Lagos (Muhammed Babayode, 2024).	routers for WAN connectivity.	traffic filtering, Port Security to prevent unauthorized access, and NAT for public/private IP mapping.	management for the faculty.
6	Perancangan Jaringan Zero Link Failure pada Layer 2 (Samuel Aras Michael & Boy Yuliadi, 2025).	Cisco Packet Tracer, GNS3 software, Wireshark	Layer 2 network design for PT. XYZ (manufacturing) integrating STP and EtherChannel to ensure continuous operation during link failures.	Zero Link Failure: Implementation of redundancy via STP (loop prevention) and EtherChannel (link aggregation). Recovery Time: Measured in seconds (target < 5s). Stability: Throughput capacity and connectivity maintenance. Rapid Recovery: The network maintained connectivity with a recovery time of under 5 seconds during physical link failure. Enhanced Stability: EtherChannel successfully aggregated links to increase throughput while STP prevented loops

7	Perancangan Jaringan Redundancy Menggunakan Konsep Etherchannel dan HSRP dengan Intervlan Routing pada PLN UID Jakarta Raya. (Octavian, 2024).	Cisco Packet Tracer	Network design for PLN UID Jakarta Raya using 3 Multilayer Switches (Core/Distribution) and 10 Access Switches to connect 10 different VLANs (Lobby, Data Center, etc.).	Redundancy & Failover: HSRP for gateway redundancy and EtherChannel for link aggregation.	Connectivity & Recovery: Successful ping replies between VLANs	Seamless Redundancy: When the primary switch (MLS-DISJAYA-2) was disabled, the standby switch (MLS-DISJAYA-3) successfully took over traffic.
8	Implementasi Konfigurasi Spanning-Tree Protocol (STP) dan Rappid Spanning-Tree Protocol (RSTP) di Cisco Packet Tracer (Anggi Silalahi et al., 2025).	Cisco Packet Tracer	A loop-prone topology using 3 Cisco 2960 Switches and 3 PCs connected via straight and crossover cables to force a redundant path selection	Convergence Speed: Comparing the time taken for port transitions (Blocking to Forwarding) between legacy STP and RSTP	Convergence Time: Measured in seconds (STP ~30s vs. RSTP ~2s) Port States: Observation of transition phases (Listening/Learning vs. Direct Forwarding).	Speed Superiority: RSTP is significantly faster, converging in a few seconds by skipping the "Listening" and "Learning" states, whereas STP takes ~30 seconds
9	Perancangan Infrastruktur Jaringan Berbasis Aplikasi Packet Tracer dengan Metode HSRP	Cisco Packet Tracer	Network infrastructure design for PT. Quantum implementing HSRP to create a redundant	Failover capability (Active to Backup router transition) and	Latency, Packet Loss	Low Latency: Average ping reply speed was consistently below 10 ms

	(Yogasetya Suhanda et al., 2022).		link (backup path) for high availability	routing path verification	(specifically 3-10 ms for active connections).
10	Comparative Analysis of Spanning Tree Protocol and Rapid Spanning Tree Protocol using Packet Tracer (Rathod, 2022).	Cisco Packet Tracer	A topology using 5 layer-2 switches connected in a ring, connecting one server and four computers	Packet Loss during Convergence: Comparing how many packets are dropped when a link fails and the network switches to an alternate path	Zero Packet Loss: The network successfully transitioned from the Active to the Standby router without losing data. RSTP Efficiency: RSTP lost only 1 packet (7% loss) during link failure, while STP lost 12 packets (53% loss) Convergence Speed: RSTP demonstrated significantly faster convergence than STP
11	Simulation and Performance Analysis of Network Backup Systems (HSRP)	Cisco Packet Tracer, GNS3	Banking sector network topology with Core (Catalyst 3560), Distribution, and Access layers,	Failover Capability: Active to Standby transition during	Downtime Duration: Time taken to restore connection. Business Continuity: HSRP successfully minimized link downtime by automatically

	(Simanjuntak et al., 2023).	software, Wireshark	comparing a single-core design vs. a redundant HSRP design	core switch failure	QoS: Throughput, Delay, Jitter, and Packet Loss.	switching to the standby router when the active device failed, preventing transaction disruptions critical for banking.
12	Performance and Availability Evaluation of an Smart Hospital Architecture (Rodrigues et al., 2021)	Stochastic Petri Nets (SPN)	Data transmission flow from Wireless Sensor Networks (WSN) to Edge and Cloud servers	Request Arrival Rate (AD) and Server Capacity	Mean Response Time (MRT), Discard Probability, and Availability	Server redundancy (Scenario A) achieved the highest availability of 99.9199% (approx. 7 hours downtime/year)
13	Simulasi dan Perancangan Redudansi Jaringan Berbasis Cisco Packet Tracer (Rasyid et al., 2024)	Cisco Packet Tracer	School network (SMKS Handayani Makassar) with 2 computer labs (16 PCs total), implementing HSRP for gateway redundancy to connect to a server via a Cloud/DSL modem.	QoS Standards: Comparison of network performance against THIPON standards across three scenarios (Normal, Failover, Recovery).	QoS Metrics: Delay (<150ms), Throughput (>2.1 Mbps), Packet Loss (%), and Jitter (ms).	High Availability: HSRP successfully maintained connection during router failure with zero packet loss in most scenarios.

Table 2.6: Literature Review Simulation Table

2.7 Summary of the Chapter

In this chapter, from what we explained above, hopefully this chapter will provide an overview of the project. This chapter also covers the previous research study that are suitable for improving the network. This chapter mainly covers the findings from previous research paper that later builds up the foundation of this project's existence.

CHAPTER 3

METHODOLOGY

3.1 Introduction

This chapter discusses the methodology used to design, implement and evaluate a resilient hospital network architecture. It highlights the approach taken to enhance the existing network design by implementing redundancy protocols to improve the network's resilience and availability. The methodology includes explanation regarding System Development Life Cycle model used for this project along with explanation of processes that happened in each phase, network and software requirements, network architecture, project framework, flowcharts, techniques and proof of concepts.

3.2 System Development Life Cycle (SDLC) Iterative Model

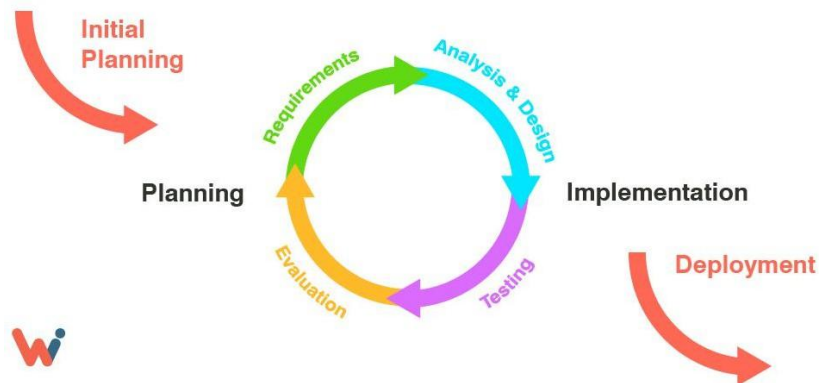


Figure 3.1: Iterative model

An iterative SDLC model, as shown in *Figure 3.1*, is chosen because of its compatibility with progressive refinement through repetitive requirement analysis, design, implementation, testing and evaluation cycle. The project begins analyzing the weakness of the existing network, which is then followed up with replicating the

network design and configuration in Cisco Packet Tracer by following the descriptions written in the paper. Then, the network is tested, and the results were noted down and evaluated. This cycle is repeated with network component failure and precedingly with the implementation of the chosen redundancy protocols.

3.2.1 Phase 1: Requirement Analysis

In this phase, a research paper is selected and is analyzed to understand the topology, protocols used, overall configuration and function. It is identified that the network lacks redundancy measures, thus the need for network performance requirements like high availability and quick failover. Network availability, failover time, packet loss and packet delivery ratio are chosen to be the performance metrics to measure the before and after redundancy implementation for further evaluation.

3.2.2 Phase 2: Network Design

The original network architecture is designed in Cisco Packet Tracer, following the same number of VLANs, Multilayer Switches (MLS), Layer 2 Access Switches, Routers, Access Points and end devices (Computer, Laptop, Printer, Smartphone) as well as protocols (DHCP, OSPF) and network services. Network design limitations or flaws were identified, particularly those related to redundancy like missing gateway redundancy mechanisms. Then, enhancements are provided by introducing redundancy protocols to the original network, thus creating a more redundant network without changing the core architecture of the existing hospital network.

3.2.3 Phase 3: Implementation

Throughout this phase, there are two stages of network implementation. First one is the

baseline network. Baseline network refers to the network designed and configured according to the paper. After testing was done on the baseline network, the redundancy protocols (HSRP, RSTP) implementation is initiated to enhance the network resilience. This is the second stage of the Implementation Phase.

3.2.4 Phase 4: Testing

Testing is conducted on both baseline and enhanced network. Performance metrics are recorded under both normal and failure scenarios. The four-performance metrics measured are availability, failover time, packet loss and PDR. Measurement for each is taken as below:

3.2.4.1 Failover Time

Failover time is the time taken for the network to switch from a failed component to a backup component. It is determined when doing a continuous ping test for 100 seconds. The period of when the ping test started showing “Request Timed Out” after a failure is triggered until it received a reply again is observed and noted down. The formula is as stated below:

$$\text{Failover Time} = T_{\text{ping resumes}} - T_{\text{failure}}$$

3.2.4.2 Availability

Availability is the time required for a network or system to remain operational and accessible to users during a specified observation period. To find availability, failover time needs to be determined first. The formula is as follows:

Where:

$$\text{Availability (\%)} = \frac{\text{Total time} - \text{Downtime}}{\text{Total Time}} \times 100\%$$

- Total Time = Ping test period (100 seconds)
- Downtime = Period of when the ping test started showing “Request Timed Out” until it received a reply again (Failover Time).

3.2.4.3 Packet Loss

Packet loss occurs when data packets traveling across a computer network fail to reach their intended destination. To find the amount of packet loss, a ping test is done between PC A with PC B. Subsequently, a failure scenario is initiated and when it back online and the ping test is halted, the amount of packet sent, and packet loss are noted down.

3.2.4.4 Packet Delivery Ratio (PDR)

PDR is a measurement of the reliability of a communication system obtained through calculating the percentage of data packets successfully received at the destination out of the total packets sent by the source. Packet Sent is set to 50 packets to ensure the Packet Delivery Ratio (PDR) provided a direct, unbiased comparison of protocol performance during the exact failover window since packet loss in this scenario only happened when a failure is initiated. The formula is as follows:

$$PDR(\%) = \left(\frac{(Packet\ Sent - Packet\ Loss)}{Packet\ Sent} \right) \times 100$$

3.2.5 Phase 5: Deployment

The enhanced network configuration is considered as the deployed system in the simulation. It represents the resilient hospital network using redundancy protocols and is served as a reference for real-world implementation and deployment.

3.2.6 Phase 6: Maintenance and Evaluation

Performance results from both baseline and enhanced network are compared and evaluated. Any improvement in the performance metrics (availability, failover time, packet loss, PDR) is further analyzed to assess the effectiveness of the implemented protocols. Observed limitations are also documented as potential areas for improvement in the future.

3.3 Network Requirements

Since this project is done purely on a simulation tool, only Laptop or Computer is needed for hardware requirements.

3.3.1 Software Requirements

Table 3.1 shows the software used throughout this project.

Tool	Descriptions
Cisco Packet Tracer v9.0.0	For network design and simulation.
Canva	For flowcharts, framework and network architecture.

Table 3.1: Software Requirements Table

3.4 Network Design

3.4.1 Network Architecture

The network uses 2-tier network architecture or Collapsed Core Design. It is modern and simplified version of the 3-tier model which consists of Core, Distribution and Access layer. In this architecture, core and distribution layers are merged forming a single layer. This saves costs and reduces network complexity, thus making troubleshooting easier. The network architecture is as shown in *Figure 3.2*.

3.4.1.1 The Collapsed Core/Distribution Layer

This layer includes the HQ and Branch Routers as well as the MLS switches

equipped with HSRP Active/Standby and EtherChannel with LACP implemented. It is also responsible for handling functions like inter-VLAN routing which allows different departments to talk to one another, redundancy as it provides default gateways through HSRP for all end devices and aggregation because it aggregates all the traffic from access layer switches and forwards them to the Internet Service Providers (ISPs) (Firdaus Syah & Adriyanto, 2025; Paguigan et al., 2022).

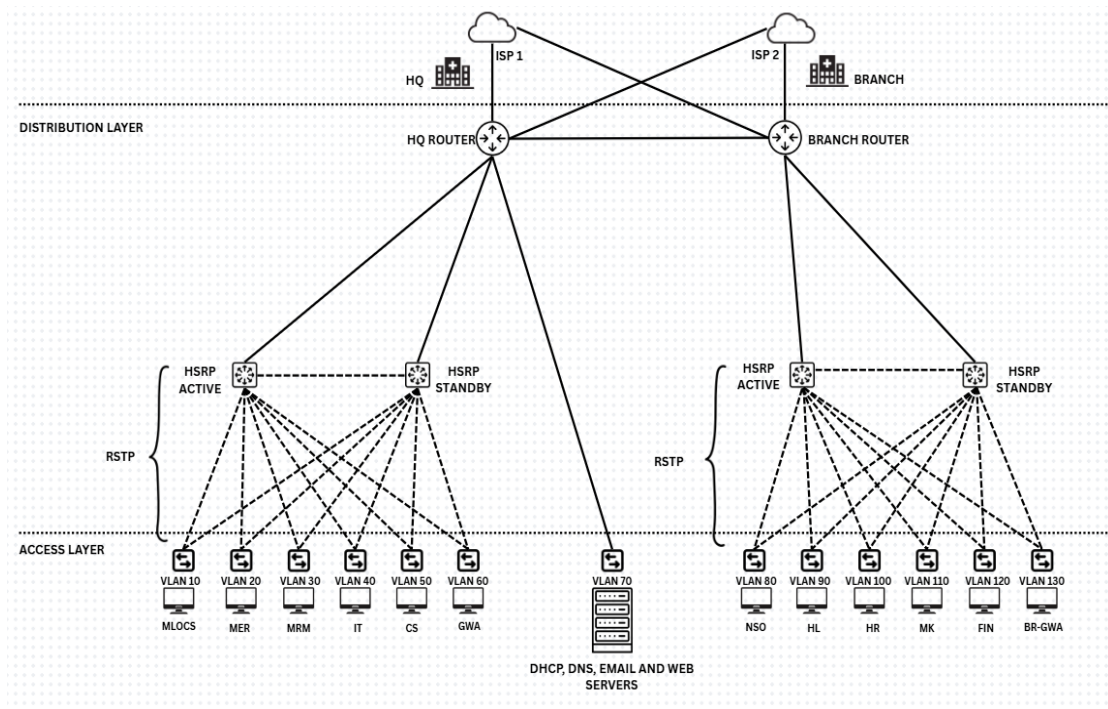


Figure 3.2: Network Architecture

3.4.1.2 The Access Layer

Access layer in this architecture represents the switches at the bottom labeled with VLANs. This layer provides connectivity because it connects end-user devices to the network, segmentation since VLAN tagging is done in this layer and uplinks as it uses RSTP to connect with Distribution Layer, thus preventing layer 2 loops while maintaining redundant paths (Rathod, 2022).

3.4.2 Project Framework

This framework highlights a methodological approach taken to validate and enhance an existing hospital network design by doing a comparative study. As shown in *Figure 3.3*, the project begins with implementing the design and configuration used by the original paper to produce a baseline network. The network is then tested to gather results before another testing with failure scenarios is initiated.

The redundancy protocols, HSRP and RSTP are then configured and another series of testing is done to obtain the results. The results are then compared and evaluated before the selected Performance Metrics, network availability, failover time, packet loss and PDR are calculated.

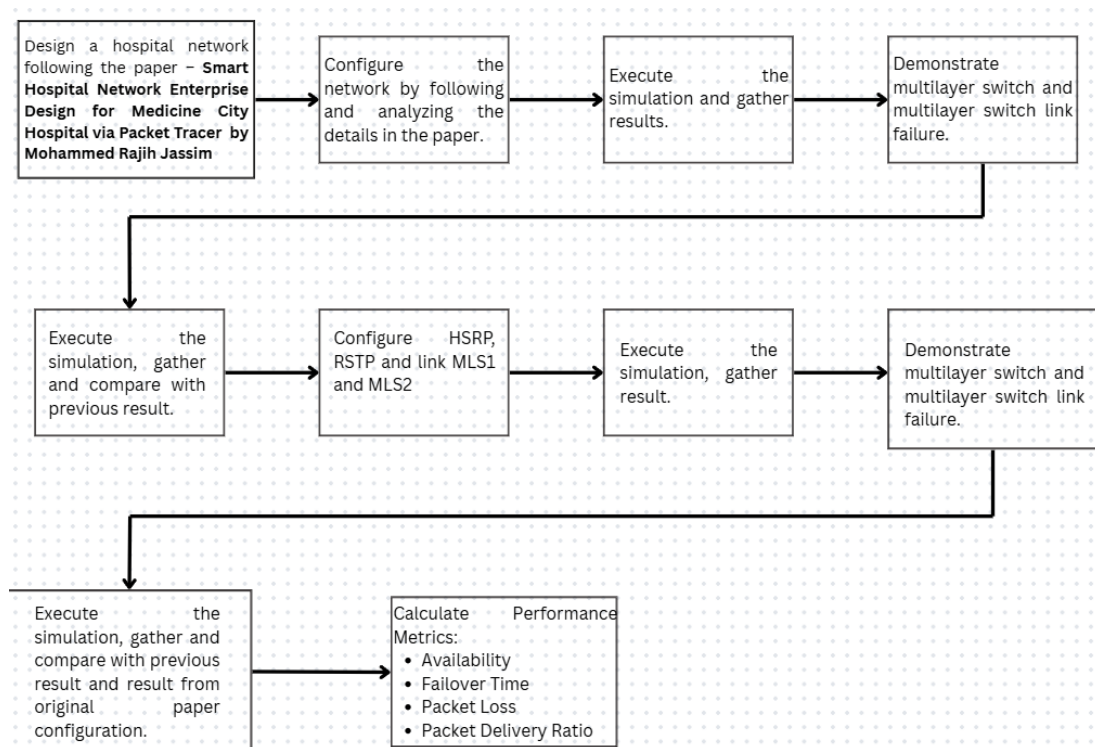


Figure 3.3: Project Framework

3.5 Project Flowchart

This flowchart maps the project methodology for enhancing the hospital network. It begins with Requirement Analysis to identify the limitations in the network proposed by the paper. Amongst the limitations identified, redundancy is chosen to be the focus as it is an important aspect in a hospital network since it requires high availability (DB Technology, 2023; Sameer Malik, 2025). The network used in the paper is then replicated in Cisco Packet Tracer followed by the configurations to create a baseline network. Several tests are done during normal and failure conditions to obtain results.

Then the techniques proposed for this project, the redundancy protocols, are implemented before another testing is done. Then the results for baseline and enhanced networks are compared and evaluated before the performance metrics are calculated. All results, performance metrics, observations and conclusions are then documented. The flow of the project is as represented in *Figure 3.4*.

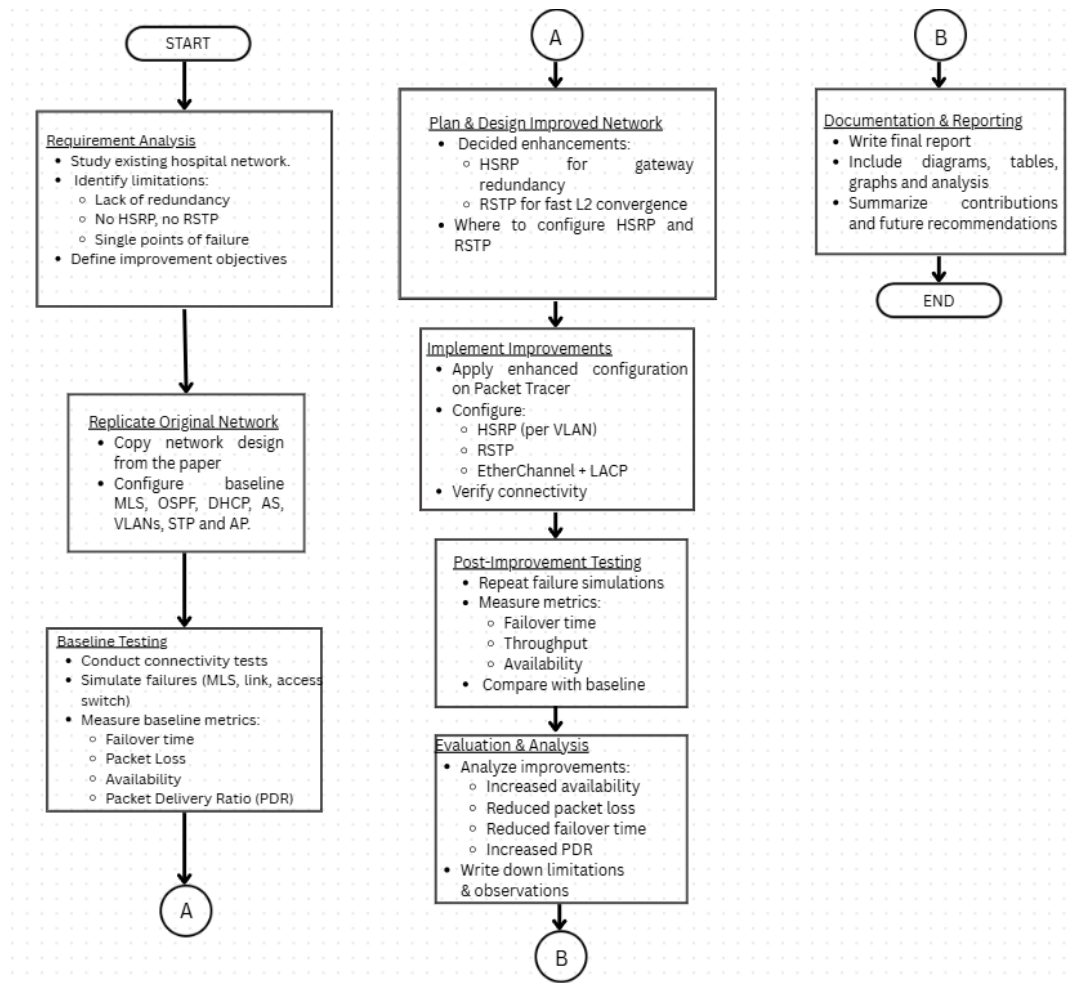


Figure 3.4: Project Flowchart

3.6 Techniques

The techniques proposed for this project are HSRP and RSTP. These protocols are deployed in the original network to produce an enhanced hospital network which is more resilient to component failures and produces better results when measured using performance metrics. A peer link is also added in between MLS1 and MLS2 which promotes seamless traffic redirection and sub-second convergence, preventing packet loss during a link failover.

3.6.1 Hot Standby Router Protocol (HSRP)

HSRP is a Cisco Proprietary protocol that allows multiple routers and switches to work together in a group, enabling default gateway redundancy and failover mechanism in the network (Octavian, 2024; Simanjuntak et al., 2023).

3.6.1.1 HSRP Active/Standby Role Assignment

MLS1 on both HQ and Branch sides are configured to be HSRP Active while MLS2 is HSRP Standby. The active switch is responsible for handling all routing traffic for all departments during normal operations while the second switch stays on Standby, which is an idle state where the switch would only monitor the active switch through hello messages. If the active switch fails completely, the standby switch will take over the active role. While this does not fully utilize the second switch during normal operation, it allows simplicity and predictability which helps in troubleshooting.

Figure 3.5 shows the flow of how HSRP active and standby role are assigned to devices.

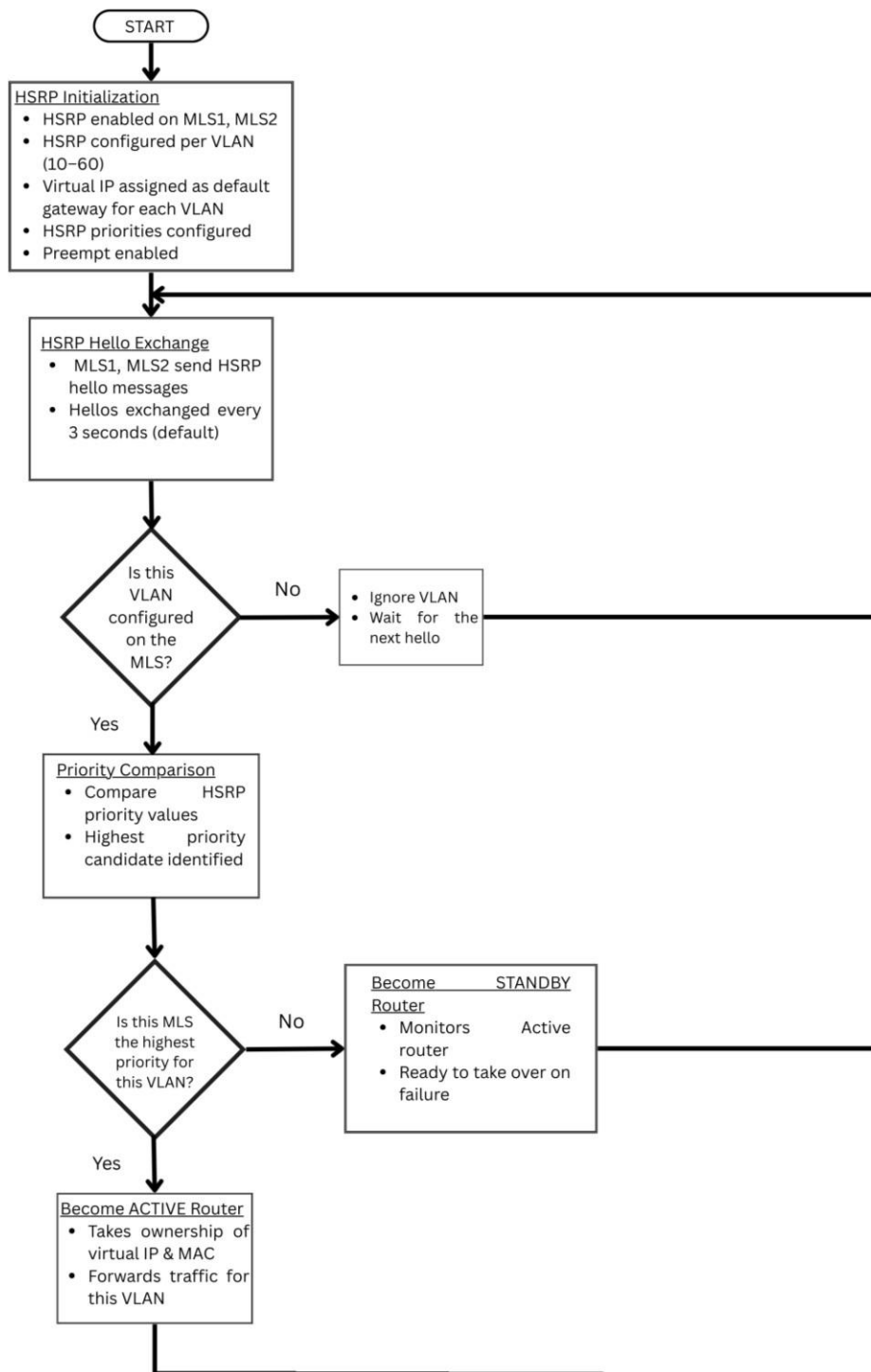


Figure 3.5: HSRP Role Assignment

3.6.1.2 HSRP Failover and Recovery Process

This process explains how HSRP failover system works and how it recovers after a failure occurs. HSRP relies on continuous exchange of hello messages in which the Standby listens, while the Active sends the messages. If an Active switch goes silent or fails, Standby's hold timer will expire, and it will immediately assume that the active switch has crashed. It will then promote itself to take on the Active role, taking over Virtual IP and MAC address as to keep the traffic flowing without showing any sign of significant outage happening to the end users.

When the previous Active switch has recovered, it will not take over the Active role instantly. It will need to check its Priority first. If preempt is configured and the switch priority is higher than the current Active router, it will have to wait for the Preempt Delay timer before it can take over the Active role again as to ensure it has fully recovered and well-stabilized. This prevents constant switching in the network from happening due to the device restarting or any cable is found loose. However, if the priority is lower, the recovered switch will return as Standby switch.

Figure 3.6 shows how HSRP failover works and how when it goes down, undergoes recovery and rejoined the network depending on its assigned priority.

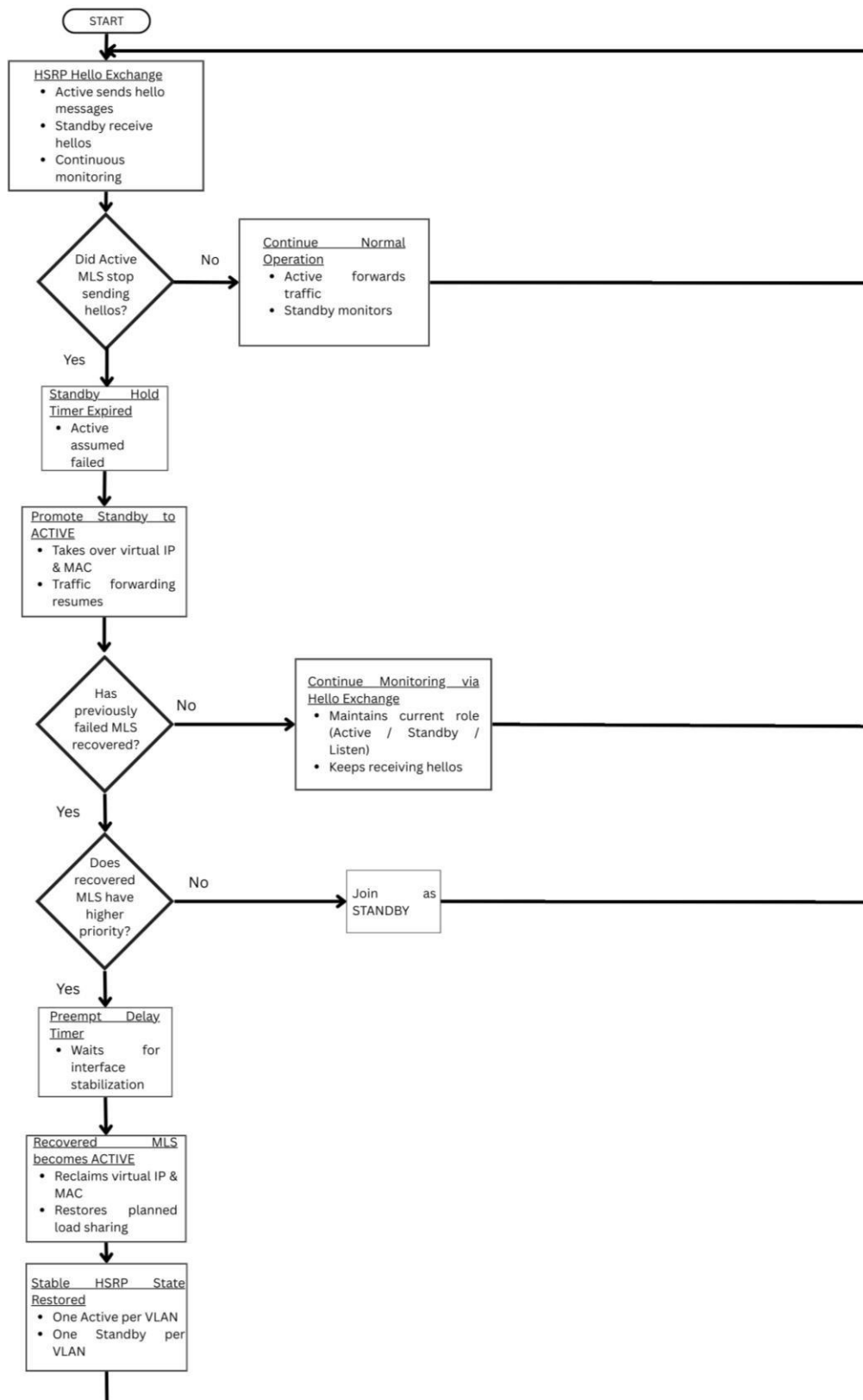


Figure 3.6: HSRP Failover + Recovery

3.6.2 Rapid Spanning Tree Protocol (RSTP)

RSTP is responsible for preventing layer 2 loops which are very destructive to a network as it leads to broadcast storms that can damage the network components. It also maintains redundant backup paths. All switches will first exchange Bridge Protocol Data Units (BPDUs) to elect a “Root Bridge” which acts as a reference point for the network. From this election, other switches will calculate the shortest path to the root, thus designating its ports as either Forwarding or Blocking. Forwarding is the active path while Blocking acts as backup paths.

After the network is free from loops, it will enter Normal Operation, which is a state where the switches continuously monitor the network status. If any changes in network topology are identified due to link failures or new switch being added, RSTP will trigger a Rapid Convergence process (happens in less than a second), which is different compared to traditional STP that waits for 30-50 seconds to do so. This immediately transitions the alternate ports to Forwarding state, thus restoring connectivity almost instantly.

Figure 3.7 shows the workflow of RSTP, its port role and state assignments and how it adapts to any topological changes.

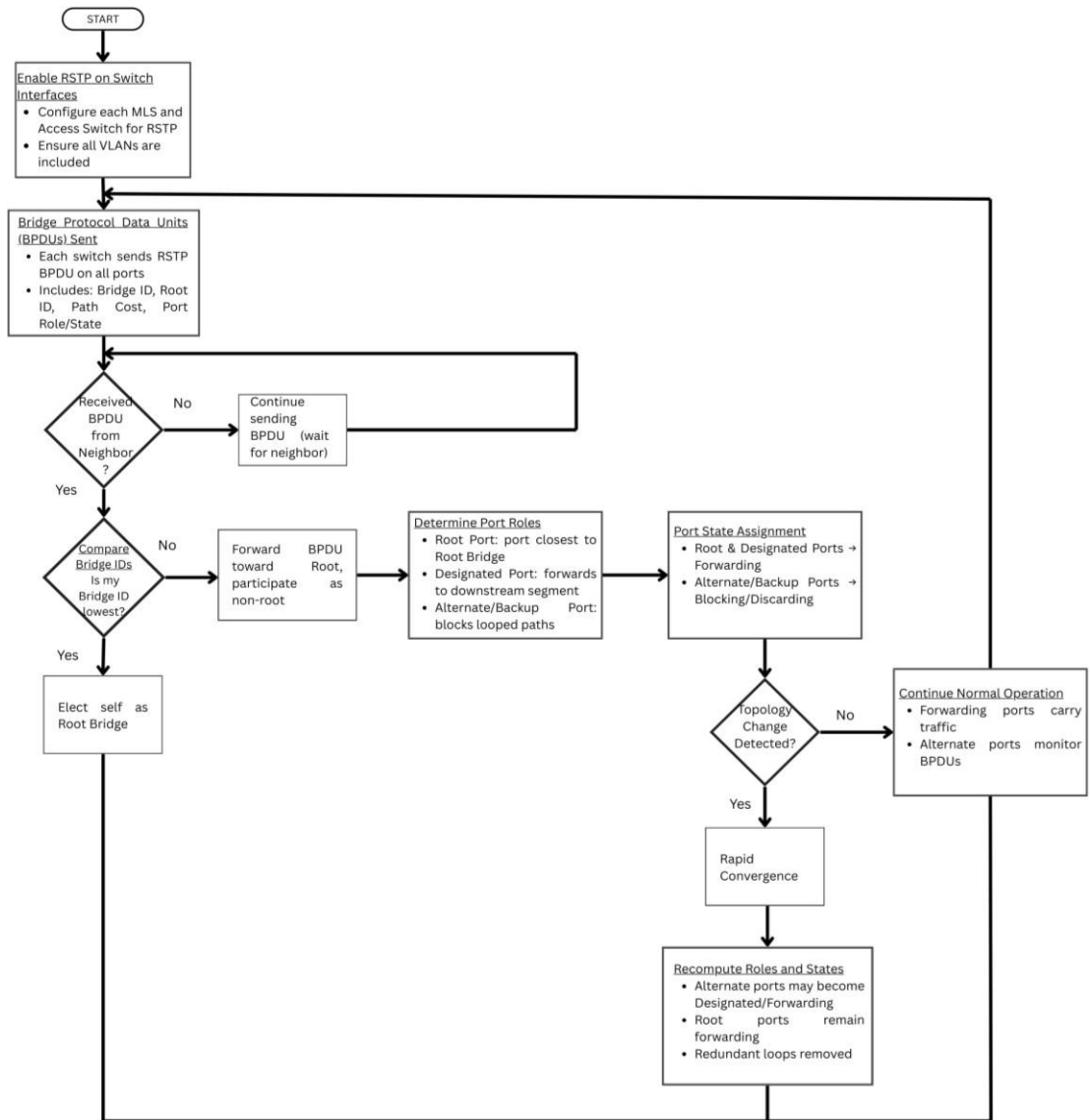


Figure 3.7: RSTP Process

3.7 Proof of Concept

3.7.1 Network Topology

Network Topology in Cisco Packet Tracer. Replicating the original network from a paper.

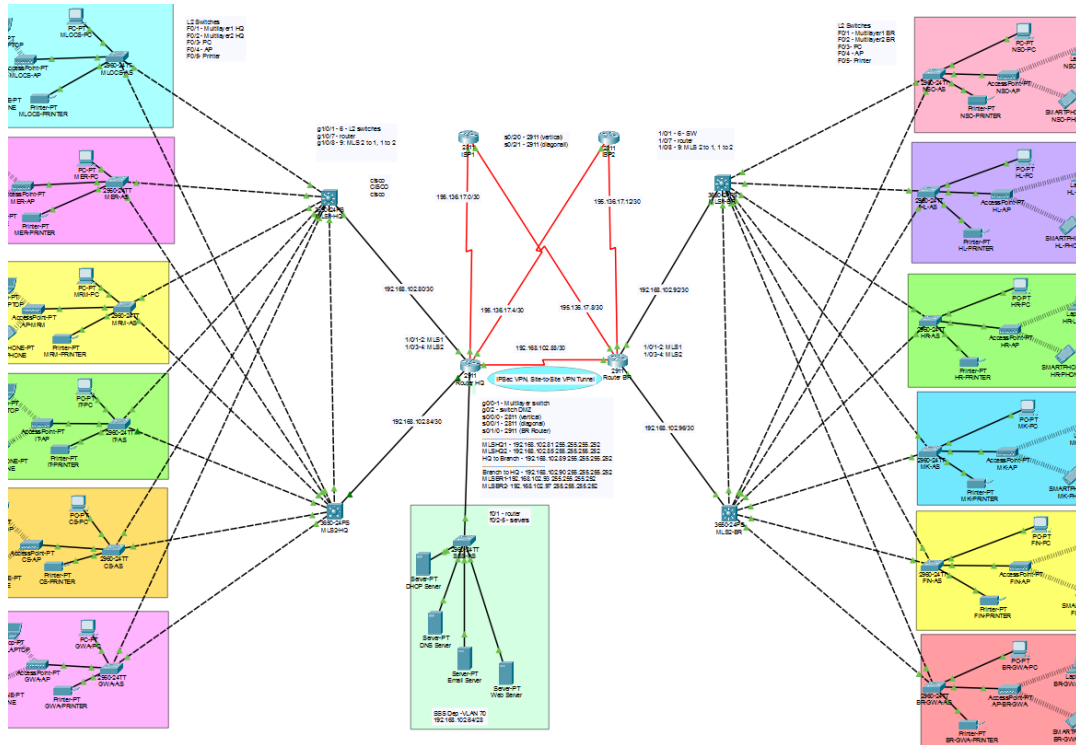


Figure 3.8: Hospital network topology

3.7.2 VLANs

For VLAN assignments, each MLS handles 6 department VLANs, 1 Management VLAN and 1 VLAN which is used to relocate unused ports for security. For Layer 2 Access switches, each handle 3 VLAN. 1 for their own department, 1 Management VLAN and 1 security VLAN.

```

AS-BR-GWA(config-if-range)#do sh vlan br

```

VLAN Name	Status	Ports
1 default	active	
99 MANAGEMENT	active	
130 BR-GWA	active	Fa0/3, Fa0/4, Fa0/5
999 Unused-Ports	active	Fa0/6, Fa0/7, Fa0/8, Fa0/9 Fa0/10, Fa0/11, Fa0/12, Fa0/13 Fa0/14, Fa0/15, Fa0/16, Fa0/17 Fa0/18, Fa0/19, Fa0/20, Fa0/21 Fa0/22, Fa0/23, Fa0/24, Gig0/1 Gig0/2
1002 fddi-default	active	
1003 token-ring-default	active	
1004 fddinet-default	active	
1005 trnet-default	active	

```

AS-BR-GWA(config-if-range)#do wr mem
Building configuration...
[OK]
AS-BR-GWA(config-if-range)#

```

Figure 3.9: VLANs in a Layer 2 Access Switch (VLAN 130, BR-GWA)

```

MLS3-HQ1(config-if-range)#do sh vlan br

```

VLAN Name	Status	Ports
1 default	active	
10 MLOCS	active	
20 MER	active	
30 MRM	active	
40 IT	active	
50 C3	active	
60 GWA	active	
99 MANAGEMENT	active	
999 Unused-Ports	active	Gig1/0/8, Gig1/0/9, Gig1/0/10, Gig1/0/11 Gig1/0/12, Gig1/0/13, Gig1/0/14, Gig1/0/15 Gig1/0/16, Gig1/0/17, Gig1/0/18, Gig1/0/19 Gig1/0/20, Gig1/0/21, Gig1/0/22, Gig1/0/23 Gig1/0/24, Gig1/1/1, Gig1/1/2, Gig1/1/3 Gig1/1/4
1002 fddi-default	active	
1003 token-ring-default	active	
1004 fddinet-default	active	
1005 trnet-default	active	

Figure 3.10: VLANs in MLS1 (HQ side)

```

MLS3-BR1(config-if-range)#
%LINK-5-CHANGED: Interface GigabitEthernet1/0/8, changed state to administratively down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0/8, changed state to down
do sh vlan br

```

VLAN Name	Status	Ports
1 default	active	
80 NSO	active	
90 HL	active	
99 MANAGEMENT	active	
100 HR	active	
110 MK	active	
120 FIN	active	
130 BR-GWA	active	
999 Unused-Ports	active	Gig1/0/8, Gig1/0/9, Gig1/0/10, Gig1/0/11 Gig1/0/12, Gig1/0/13, Gig1/0/14, Gig1/0/15 Gig1/0/16, Gig1/0/17, Gig1/0/18, Gig1/0/19 Gig1/0/20, Gig1/0/21, Gig1/0/22, Gig1/0/23 Gig1/0/24, Gig1/1/1, Gig1/1/2, Gig1/1/3 Gig1/1/4
1002 fddi-default	active	
1003 token-ring-default	active	
1004 fddinet-default	active	
1005 trnet-default	active	

```

MLS3-BR1(config-if-range)#exit

```

Figure 3.11: VLANs in MLS1 (Branch side)

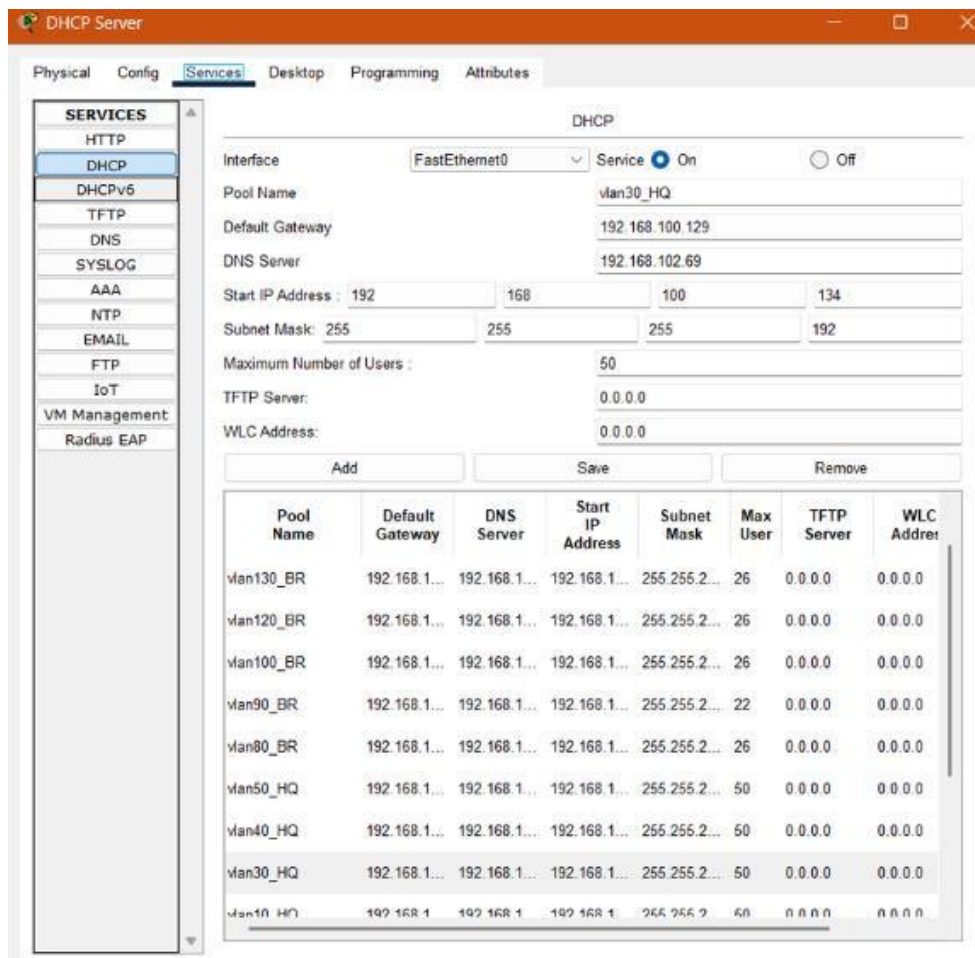


Figure 3.12: DHCP Pools created on DHCP server

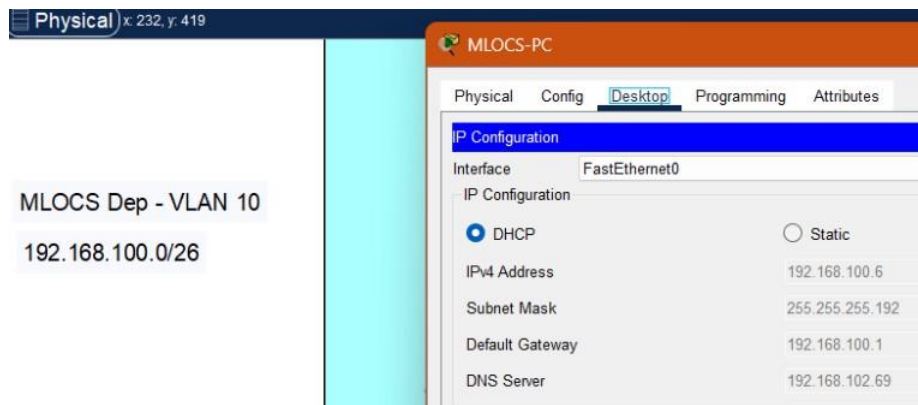


Figure 3.13: Successful DHCP in HQ VLAN



Figure 3.14: Successful DHCP in Branch VLAN

3.7.3 Testing Network Connectivity through Ping Test

Ping test are done to ensure inter-VLAN routing and inter-hospital routing are successful.

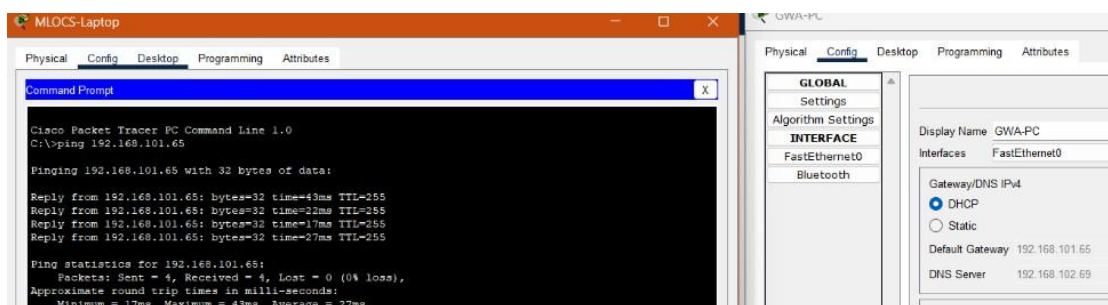


Figure 3.15: Successful inter-VLAN routing on HQ side

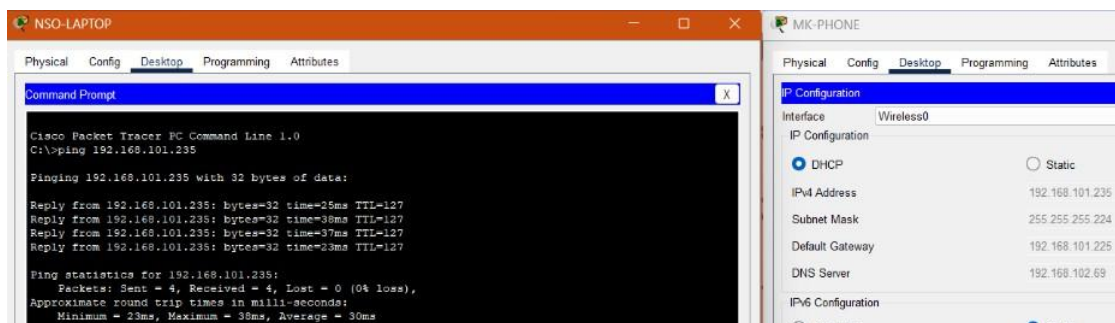


Figure 3.16: Successful inter-VLAN routing on Branch side

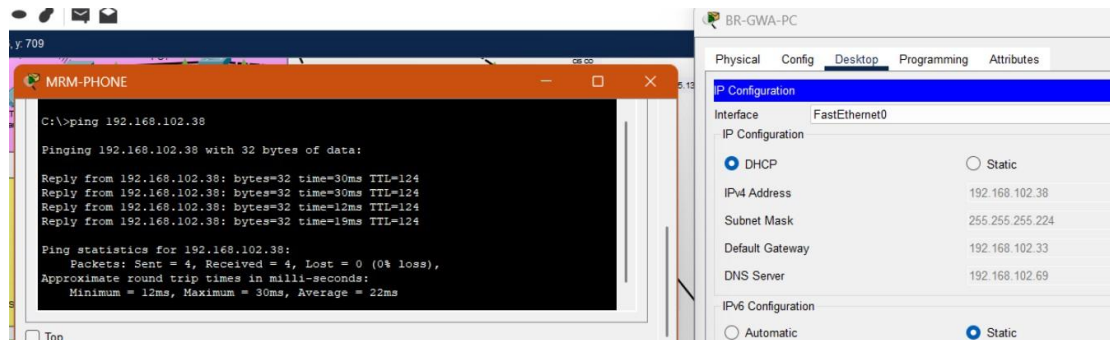


Figure 3.17: Successful inter-VLAN routing between VLAN on HQ side with VLAN on Branch side

3.7.4 OSPF Configuration.

```
HQ-ROUTER#sh ip ospf neighbor
```

Neighbor ID	Pri	State	Dead Time	Address	Interface
10.10.10.1	1	FULL/DR	00:00:38	192.168.102.82	GigabitEthernet0/0
10.10.10.2	1	FULL/DR	00:00:38	192.168.102.86	GigabitEthernet0/1
2.2.2.2	0	FULL/-	00:00:35	192.168.102.90	Serial0/1/0

```
HQ-ROUTER#sh ip ospf br
^
% Invalid input detected at '^' marker.

HQ-ROUTER#sh ip ospf int br
```

Interface	PID	Area	IP Address/Mask	Cost	State	Nbrs	F/C
Gig0/0	1	0	192.168.102.81/255.255.255.252	1	BDR	0/0	
Gig0/1	1	0	192.168.102.85/255.255.255.252	1	BDR	0/0	
Gig0/2	1	0	192.168.102.65/255.255.255.240	1	DR	0/0	
Se0/1/0	1	0	192.168.102.89/255.255.255.252	64	POINT	0/0	

```
HQ-ROUTER#sh ip ospf int vlan
^
% Invalid input detected at '^' marker.

HQ-ROUTER#show ip route ospf
```

```

192.168.99.0/24 is variably subnetted, 2 subnets, 2 masks
O   192.168.99.0 [110/2] via 192.168.102.82, 00:15:43, GigabitEthernet0/0
    [110/2] via 192.168.102.86, 00:15:43, GigabitEthernet0/1
O   192.168.99.64 [110/66] via 192.168.102.90, 00:15:33, Serial0/1/0
192.168.100.0/26 is subnetted, 4 subnets
O   192.168.100.0 [110/2] via 192.168.102.82, 00:15:43, GigabitEthernet0/0
    [110/2] via 192.168.102.86, 00:15:43, GigabitEthernet0/1
O   192.168.100.64 [110/2] via 192.168.102.82, 00:15:43, GigabitEthernet0/0
    [110/2] via 192.168.102.86, 00:15:43, GigabitEthernet0/1
O   192.168.100.128 [110/2] via 192.168.102.82, 00:15:43, GigabitEthernet0/0
    [110/2] via 192.168.102.86, 00:15:43, GigabitEthernet0/1
O   192.168.100.192 [110/2] via 192.168.102.82, 00:15:43, GigabitEthernet0/0
    [110/2] via 192.168.102.86, 00:15:43, GigabitEthernet0/1
192.168.101.0/24 is variably subnetted, 6 subnets, 2 masks
O   192.168.101.0 [110/2] via 192.168.102.82, 00:15:43, GigabitEthernet0/0
    [110/2] via 192.168.102.86, 00:15:43, GigabitEthernet0/1
O   192.168.101.64 [110/2] via 192.168.102.82, 00:15:43, GigabitEthernet0/0
    [110/2] via 192.168.102.86, 00:15:43, GigabitEthernet0/1
O   192.168.101.128 [110/66] via 192.168.102.90, 00:15:43, Serial0/1/0
O   192.168.101.160 [110/66] via 192.168.102.90, 00:15:43, Serial0/1/0
O   192.168.101.192 [110/66] via 192.168.102.90, 00:15:43, Serial0/1/0
O   192.168.101.224 [110/66] via 192.168.102.90, 00:15:43, Serial0/1/0
192.168.102.0/24 is variably subnetted, 12 subnets, 4 masks
O   192.168.102.0 [110/66] via 192.168.102.90, 00:15:43, Serial0/1/0
O   192.168.102.32 [110/66] via 192.168.102.90, 00:15:43, Serial0/1/0
O   192.168.102.96 [110/65] via 192.168.102.90, 00:15:43, Serial0/1/0
O   192.168.102.96 [110/65] via 192.168.102.90, 00:15:43, Serial0/1/0
```

Figure 3.18: OSPF Configuration on HQ Router

```

BRRouter>en
BRRouter#sh ip ospf neighbor

Neighbor ID      Pri   State           Dead Time   Address        Interface
30.30.30.1       1     FULL/DR         00:00:34    192.168.102.94 GigabitEthernet0/0
30.30.30.2       1     FULL/DR         00:00:34    192.168.102.98 GigabitEthernet0/1
1.1.1.1          0     FULL/-          00:00:35    192.168.102.89 Serial0/1/0

BRRouter#sh ip ospf int vian
^
% Invalid input detected at '^' marker.

BRRouter#sh ip ospf int br
Interface        PID   Area           IP Address/Mask    Cost   State   Nbrs  F/C
Gig0/0           1     0              192.168.102.93/255.255.255.252  1      BDR    0/0
Gig0/1           1     0              192.168.102.97/255.255.255.252  1      BDR    0/0
Se0/1/0          1     0              192.168.102.90/255.255.255.252  64     POINT  0/0

BRRouter##show ip route ospf
^
% Invalid input detected at '^' marker.

BRRouter#show ip route ospf
      192.168.99.0/24 is variably subnetted, 2 subnets, 2 masks
O       192.168.99.0 [110/66] via 192.168.102.89, 00:23:13, Serial0/1/0
O       192.168.99.64 [110/2] via 192.168.102.94, 00:23:13, GigabitEthernet0/0
          [110/2] via 192.168.102.98, 00:23:13, GigabitEthernet0/1
      192.168.100.0/26 is subnetted, 4 subnets
O       192.168.100.0 [110/66] via 192.168.102.89, 00:23:13, Serial0/1/0
O       192.168.100.64 [110/66] via 192.168.102.89, 00:23:13, Serial0/1/0
O       192.168.100.128 [110/66] via 192.168.102.89, 00:23:13, Serial0/1/0
O       192.168.100.192 [110/66] via 192.168.102.89, 00:23:13, Serial0/1/0
      192.168.101.0/24 is variably subnetted, 6 subnets, 2 masks
O       192.168.101.0 [110/66] via 192.168.102.89, 00:23:13, Serial0/1/0
O       192.168.101.64 [110/66] via 192.168.102.89, 00:23:13, Serial0/1/0
O       192.168.101.128 [110/2] via 192.168.102.94, 00:23:13, GigabitEthernet0/0
          [110/2] via 192.168.102.98, 00:23:13, GigabitEthernet0/1
O       192.168.101.160 [110/2] via 192.168.102.94, 00:23:13, GigabitEthernet0/0
          [110/2] via 192.168.102.98, 00:23:13, GigabitEthernet0/1
O       192.168.101.192 [110/2] via 192.168.102.94, 00:23:13, GigabitEthernet0/0
          [110/2] via 192.168.102.98, 00:23:13, GigabitEthernet0/1
O       192.168.101.224 [110/2] via 192.168.102.94, 00:23:13, GigabitEthernet0/0
          [110/2] via 192.168.102.98, 00:23:13, GigabitEthernet0/1
      192.168.102.0/24 is variably subnetted, 11 subnets, 4 masks
O       192.168.102.0 [110/2] via 192.168.102.94, 00:23:13, GigabitEthernet0/0
          [110/2] via 192.168.102.98, 00:23:13, GigabitEthernet0/1
O       192.168.102.32 [110/2] via 192.168.102.94, 00:23:13, GigabitEthernet0/0
          [110/2] via 192.168.102.98, 00:23:13, GigabitEthernet0/1
O       192.168.102.64 [110/65] via 192.168.102.89, 00:23:53, Serial0/1/0

```

Figure 3.19: OSPF Configuration on Branch Router

```

MLS3-HQ1#sh ip ospf neighbor

Neighbor ID      Pri   State           Dead Time   Address        Interface
1.1.1.1          1     FULL/BDR        00:00:39    192.168.102.81 GigabitEthernet1/0/7
10.10.10.2       1     FULL/DR         00:00:39    192.168.99.2   Vlan99

MLS3-HQ1#sh ip ospf int br
Interface        PID   Area           IP Address/Mask    Cost   State   Nbrs  F/C
Vlan             1     0              192.168.100.1/255.255.255.192  1      DR     0/0
Vlan             1     0              192.168.99.1/255.255.255.192  1      BDR    0/0
Vlan             1     0              192.168.100.65/255.255.255.192  1      DR     0/0
Vlan             1     0              192.168.100.129/255.255.255.192  1      DR     0/0
Vlan             1     0              192.168.100.193/255.255.255.192  1      DR     0/0
Vlan             1     0              192.168.101.1/255.255.255.192  1      DR     0/0
Vlan             1     0              192.168.101.65/255.255.255.192  1      DR     0/0
Gig1/0/7         1     0              192.168.102.82/255.255.255.252  1      DR     0/0

MLS3-HQ1#sh ip route ospf
      192.168.99.0/24 is variably subnetted, 2 subnets, 2 masks
O       192.168.99.64 [110/67] via 192.168.102.81, 00:20:30, GigabitEthernet1/0/7
      192.168.101.0/24 is variably subnetted, 6 subnets, 2 masks
O       192.168.101.128 [110/67] via 192.168.102.81, 00:20:30, GigabitEthernet1/0/7
O       192.168.101.160 [110/67] via 192.168.102.81, 00:20:30, GigabitEthernet1/0/7
O       192.168.101.192 [110/67] via 192.168.102.81, 00:20:30, GigabitEthernet1/0/7
O       192.168.101.224 [110/67] via 192.168.102.81, 00:20:30, GigabitEthernet1/0/7
      192.168.102.0/24 is variably subnetted, 8 subnets, 3 masks
O       192.168.102.0 [110/67] via 192.168.102.81, 00:20:30, GigabitEthernet1/0/7
O       192.168.102.32 [110/67] via 192.168.102.81, 00:20:30, GigabitEthernet1/0/7
O       192.168.102.64 [110/2] via 192.168.102.81, 00:20:30, GigabitEthernet1/0/7
O       192.168.102.84 [110/2] via 192.168.99.2, 00:20:30, Vlan99
          [110/2] via 192.168.102.81, 00:20:30, GigabitEthernet1/0/7
O       192.168.102.88 [110/65] via 192.168.102.81, 00:20:30, GigabitEthernet1/0/7
O       192.168.102.92 [110/66] via 192.168.102.81, 00:20:30, GigabitEthernet1/0/7
O       192.168.102.96 [110/66] via 192.168.102.81, 00:20:30, GigabitEthernet1/0/7
O*E2 0.0.0.0/0 [110/11] via 192.168.102.81, 00:20:30, GigabitEthernet1/0/7

```

Figure 3.20: OSPF Configuration on HQ MLS

```

MLS3-BR1#sh ip ospf neighbor
Neighbor ID      Pri   State           Dead Time   Address        Interface
30.30.30.2       1     FULL/DR         00:00:33    192.168.99.66  Vlan99
2.2.2.2          1     FULL/BDR        00:00:32    192.168.102.93 GigabitEthernet1/0/7

MLS3-BR1#sh ip ospf int br
Interface        PID   Area
Vlan             1     0
Vlan             1     0
Vlan             1     0
Vlan             1     0
Vlan             1     0
Vlan             1     0
Vlan             1     0
Vlan             1     0
Vlan             1     0
Vlan             1     0
Gig1/0/7         1     0

IP Address/Mask      Cost  State  Mbrs F/C
192.168.101.129/255.255.255.224  1     DR    0/0
192.168.99.65/255.255.255.224    1     BDR   0/0
192.168.101.161/255.255.255.224  1     DR    0/0
192.168.101.193/255.255.255.224  1     DR    0/0
192.168.101.225/255.255.255.224  1     DR    0/0
192.168.102.1/255.255.255.224    1     DR    0/0
192.168.102.33/255.255.255.224   1     DR    0/0
192.168.102.94/255.255.255.252   1     DR    0/0

MLS3-BR1#sh ip ospf route
^
% Invalid input detected at '^' marker.

MLS3-BR1#sh ip route ospf
192.168.99.0/24 is variably subnetted, 2 subnets, 2 masks
O   192.168.99.0 [110/67] via 192.168.102.93, 00:25:00, GigabitEthernet1/0/7
O   192.168.100.0/26 is subnetted, 4 subnets
O   192.168.100.0 [110/67] via 192.168.102.93, 00:25:00, GigabitEthernet1/0/7
O   192.168.100.64 [110/67] via 192.168.102.93, 00:25:00, GigabitEthernet1/0/7
O   192.168.100.128 [110/67] via 192.168.102.93, 00:25:00, GigabitEthernet1/0/7
O   192.168.100.192 [110/67] via 192.168.102.93, 00:25:00, GigabitEthernet1/0/7
O   192.168.101.0/24 is variably subnetted, 6 subnets, 2 masks
O   192.168.101.0 [110/67] via 192.168.102.93, 00:25:00, GigabitEthernet1/0/7
O   192.168.101.64 [110/67] via 192.168.102.93, 00:25:00, GigabitEthernet1/0/7
O   192.168.102.0/24 is variably subnetted, 8 subnets, 3 masks
O   192.168.102.64 [110/66] via 192.168.102.93, 00:25:00, GigabitEthernet1/0/7
O   192.168.102.80 [110/66] via 192.168.102.93, 00:25:00, GigabitEthernet1/0/7
O   192.168.102.84 [110/66] via 192.168.102.93, 00:24:50, GigabitEthernet1/0/7
O   192.168.102.88 [110/65] via 192.168.102.93, 00:25:00, GigabitEthernet1/0/7
O   192.168.102.96 [110/2] via 192.168.99.66, 00:25:00, Vlan99
O   [110/2] via 192.168.102.93, 00:25:00, GigabitEthernet1/0/7
O*E2 0.0.0.0/0 [110/1] via 192.168.102.93, 00:25:00, GigabitEthernet1/0/7

```

Figure 3.21: OSPF Configuration on Branch MLS

3.8 Summary of the Chapter

This chapter outlined the methodology used to design, implement and evaluate the resilient hospital network. An iterative SDLC model was adopted to support progressive refinement across six phases: requirement analysis, network design, implementation, testing, deployment, and maintenance and evaluation. The network was designed and simulated using Cisco Packet Tracer 9.0.0, adopting a two-tier collapsed-core architecture comprising a distribution layer with two Multilayer Switches (MLS1 and MLS2) and an access layer of departmental VLAN switches across both HQ and Branch sites.

Two redundancy techniques were proposed: HSRP for default gateway redundancy and RSTP for fast Layer 2 convergence and loop prevention. A peer link between MLS1 and MLS2 was also introduced to provide a direct Layer 2 transit path. Four performance metrics were defined to evaluate network resilience before and after redundancy implementation: availability, failover time, packet loss and Packet Delivery Ratio (PDR),

with their respective measurement formulas established. The proof of concept confirmed that the baseline network was fully operational, with successful DHCP assignment, inter-VLAN routing, and OSPF configuration verified across both HQ and Branch sides prior to redundancy implementation.

CHAPTER 4

IMPLEMENTATION AND RESULT

4.1 Introduction

This chapter presents the implementation and evaluation of the proposed hospital network enhancement. It is divided into two main parts. The first part covers the implementation, documenting the replication of the baseline network and the subsequent configuration of the redundancy protocols introduced in this project. The second part presents the results obtained from testing both the baseline and enhanced networks under normal operating conditions and controlled failure scenarios. Performance metrics: failover time, availability, packet loss and Packet Delivery Ratio (PDR) are measured and compared to assess the effectiveness of the implemented redundancy mechanisms.

4.2 Baseline Network

The baseline network is a replication of the hospital network design originally proposed by Jassim (2024) for Medicine City Hospital, reconstructed in Cisco Packet Tracer version 9.0.0. It serves as the reference point for this project, representing the state of the network before any redundancy enhancements are applied. The purpose of establishing the baseline is to obtain reference performance values under both normal and failure conditions, which are later compared against the enhanced network.

The network adopts a two-tier collapsed-core architecture. The distribution layer houses two Multilayer Switches (MLS1 and MLS2) on both the Headquarters (HQ) and Branch sides, connected to their respective routers and ISP links. The access layer consists of

Layer 2 Access Switches, each assigned to a specific departmental VLAN. The network comprises 13 VLANs in total: six on the HQ side (VLAN 10–60), six on the Branch side (VLAN 80–130) and one server VLAN (VLAN 70) hosting DHCP, DNS, Email and Web servers.

The baseline is pre-configured with STP, OSPF for dynamic routing, DHCP for IP address assignment, SSH for remote management, ACLs combined with NAT and PAT for security and address management, and an IPsec Site-to-Site VPN linking the HQ and Branch routers. Critically, no gateway mechanisms are present — there is no HSRP configured and no peer link between MLS1 and MLS2, making the network vulnerable to single points of failure. The topological design is as shown in *Figure 4.1*.

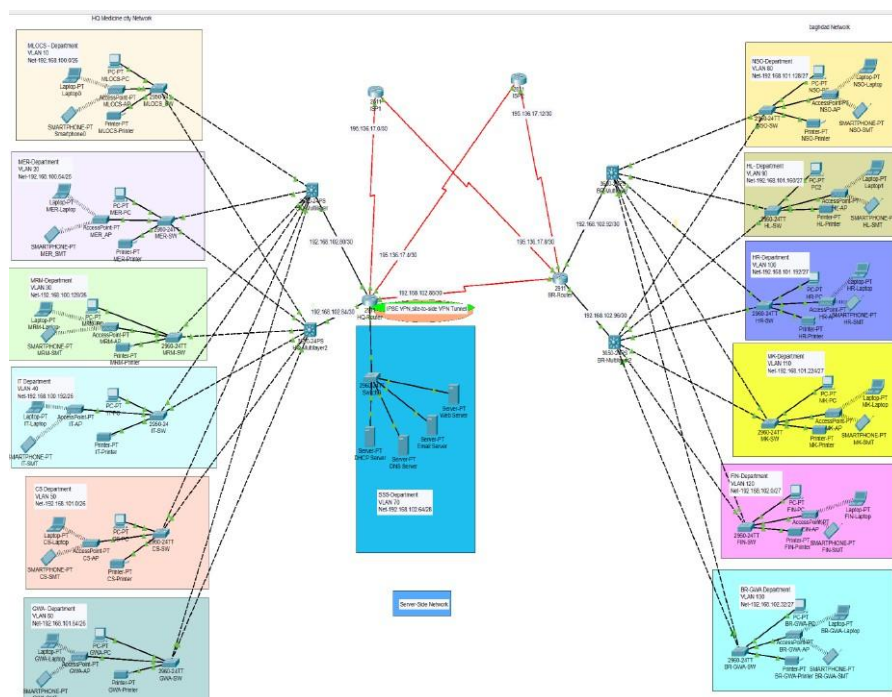


Figure 4.1: Design of a hospital network from Smart Hospital Network Enterprise

Design for Medicine City Hospital via Packet Tracer by Mohammed Rajih Jassim

(Jassim, 2024)

Table 0.1 summarizes the original network parameters and details.

Parameters	Detail
Simulation Tool	Cisco Packet Tracer 9.0.0
HQ	6 VLANs : 10 – 60 6 Access Points (1 per VLAN) 6 PC (1 per VLAN) 6 Phone (1 per VLAN) 6 Printer (1 per VLAN) 6 Laptop (1 per VLAN)
Branch	6 VLANs : 80 – 130 6 Access Points (1 per VLAN) 6 PC (1 per VLAN) 6 Phone (1 per VLAN) 6 Printer (1 per VLAN) 6 Laptop (1 per VLAN)
Distribution Layer	2 Multilayer Switches per hospital (MLS1, MLS2)
Access Layer	6 Access Switches per campus, one connecting to all servers
Servers	DHCP Server, Email Server, Web Server, DNS Server
Protocols Implemented	OSPF, DHCP, STP, SSH, ACL, NAT, PAT, IPSec Site-to-Site VPN

Table 0.2: Baseline Environment Summary

Table 0.2 shows information related to IP Addressing for both HQ and Branch sides.

Includes network address, subnet mask. Host address range and broadcast address.

HQ				
Department	Network Address	Subnet Mask	Host Address Range	Broadcast Address
MLOCS	192.168.100.0	255.255.255.192/26	192.168.100.1 to 192.168.100.62	192.168.100.63
MER	192.168.100.64	255.255.255.192/26	192.168.100.65 to 192.168.100.126	192.168.100.127
MRM	192.168.100.128	255.255.255.192/26	192.168.100.129 to 192.168.100.190	192.168.100.191
IT	192.168.100.192	255.255.255.192/26	192.168.100.193 to 192.168.100.254	192.168.100.255
CS	192.168.101.0	255.255.255.192/26	192.168.100.1 to 192.168.100.62	192.168.100.63
GWA	192.168.101.64	255.255.255.192/26	192.168.100.65 to 192.168.100.126	192.168.100.127
Branch				
Department	Network Address	Subnet Mask	Host Address Range	Broadcast Address
NSO	192.168.101.128	255.255.255.224/27	192.168.101.129 to 192.168.101.158	192.168.101.159
HL	192.168.101.160	255.255.255.224/27	192.168.101.161 to 192.168.101.190	192.168.101.191
HR	192.168.101.192	255.255.255.224/27	192.168.101.193 to 192.168.101.222	192.168.101.223
MK	192.168.101.224	255.255.255.224/27	192.168.101.225 to 192.168.101.254	192.168.101.255
FIN	192.168.102.0	255.255.255.224/27	192.168.102.1 to 192.168.102.30	192.168.102.31
GWA	192.168.102.32	255.255.255.224/27	192.168.102.33 to 192.168.102.62	192.168.102.63
Server				
Department	Network Address	Subnet Mask	Host Address Range	Broadcast Address
SSS	192.168.102.64	255.255.255.240/28	192.168.102.65 to 192.168.102.78	192.168.102.79

Table 0.2: IP Addressing

Before testing under failure conditions, the baseline network was verified to be fully functional. Connectivity tests confirmed successful DHCP assignment across all VLANs, correct inter-VLAN routing on both the HQ and Branch sides, and successful inter-site communication between HQ and Branch VLANs.

Figure 4.2 shows DHCP pools created in DHCP server for automatic IP assignments to devices in each department.

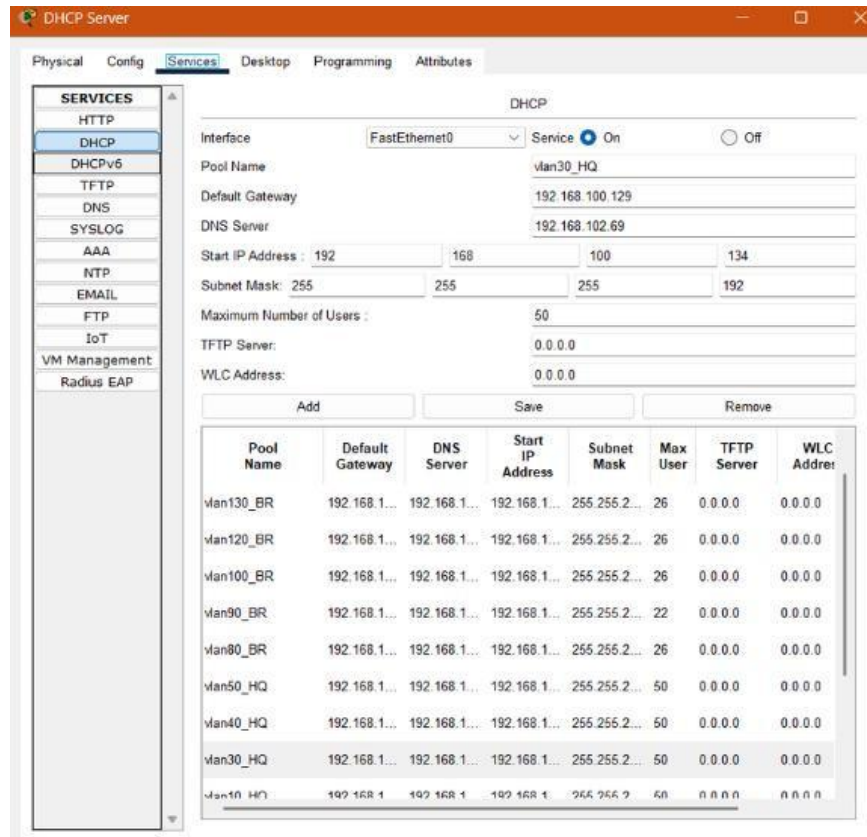


Figure 4.2: DHCP Pools created on DHCP server

Figure 4.3 shows successful IP assignment in VLAN 10 MLOCS Department (HQ) using DHCP. Can confirmed as 192.168.100.6 is within the usable address range of 192.168.100.0/26.

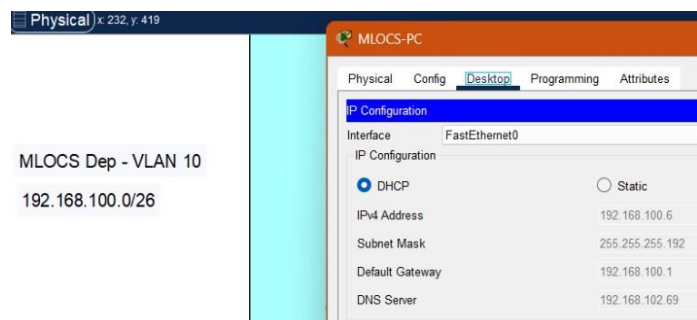


Figure 4.3: Successful DHCP in HQ VLAN

Figure 4.4 shows successful IP assignment in VLAN 80 NSO Department (Branch) using DHCP. Can confirmed as 192.168.101.135 is within the usable address range of 192.168.101.128/27.



Figure 4.4: Successful DHCP in Branch VLAN

Figure 4.5 shows successful inter-VLAN routing between 2 HQ departments (MLOCS-GWA). Can confirmed as MLOCS-Laptop received replies from GWA-PC.

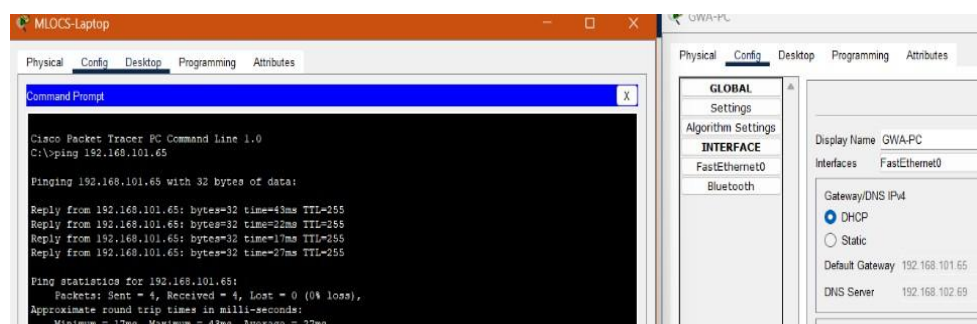


Figure 4.5: Successful inter-VLAN routing on HQ side

Figure 4.6 shows successful inter-VLAN routing between 2 Branch departments (MLOCS-GWA). Can confirmed as NSO-Laptop received replies from MK-Phone.

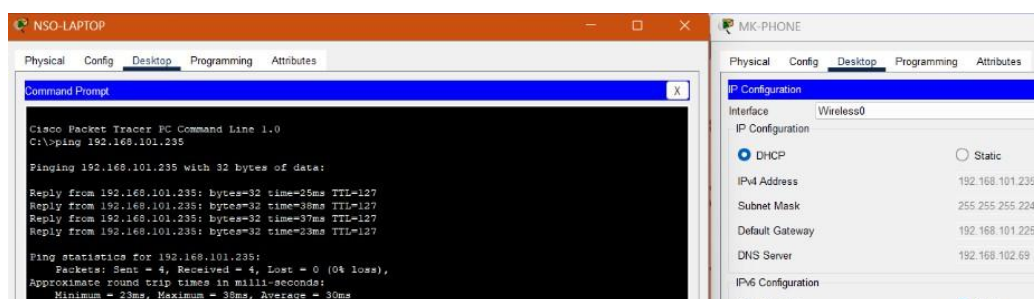


Figure 4.6: Successful inter-VLAN routing on Branch side

Figure 4.7 shows successful inter-VLAN routing between Department in HQ (MRM) and Department in Branch (BR-GWA). Can confirmed as MRM-Phone received replies from BR-GWA-PC.

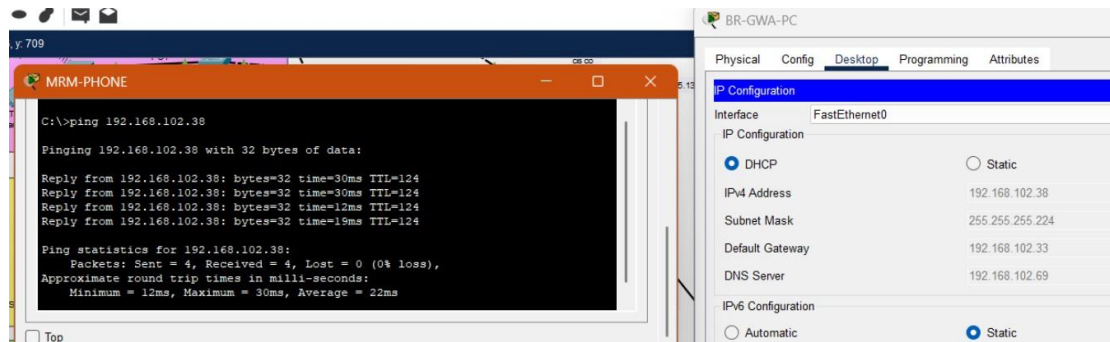


Figure 4.7: Successful inter-VLAN routing between VLAN on HQ, Branch

4.3 Enhanced Network Implementation

The enhanced network, shown in *Figure 4.8*, is produced by augmenting the baseline with two redundancy mechanisms: Hot Standby Router Protocol (HSRP) for default gateway redundancy, and Rapid Spanning Tree Protocol (RSTP) with an inter-MLS peer link. All existing baseline configurations — STP, OSPF, DHCP, SSH, ACLs, NAT, PAT and IPsec VPN — are retained without modification. The enhancements are applied to both the HQ and Branch sides of the network.

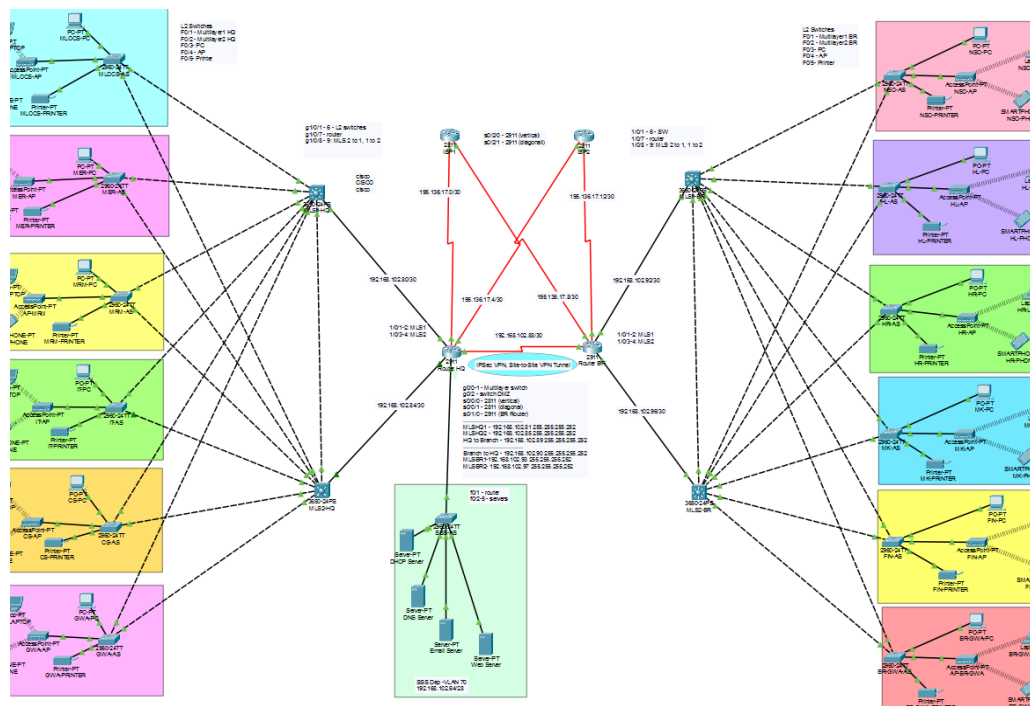
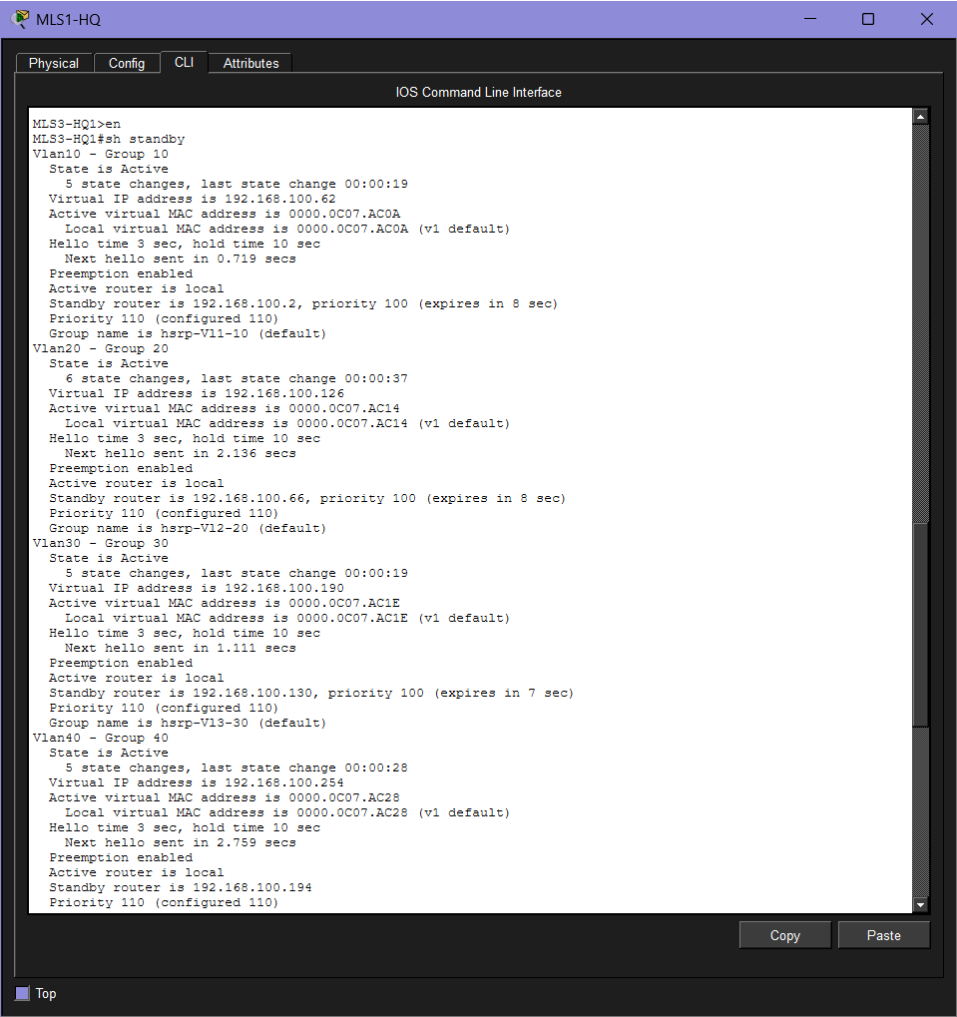


Figure 4.8: Enhanced hospital network topology with redundancy mechanisms applied

4.3.1 HSRP Configuration

HSRP is configured on a per-VLAN basis on MLS1 and MLS2 on both the HQ and Branch sides. MLS1 is assigned a higher HSRP priority for all VLANs to ensure it assumes the Active role under normal operating conditions as shown in *Figure 4.9*.



```
MLS1-HQ1>en
MLS1-HQ1#sh standby
Vlan10 - Group 10
  State is Active
    5 state changes, last state change 00:00:19
  Virtual IP address is 192.168.100.62
  Active virtual MAC address is 0000.0C07.AC0A
  Local virtual MAC address is 0000.0C07.AC0A (v1 default)
  Hello time 3 sec, hold time 10 sec
  Next hello sent in 0.719 secs
  Preemption enabled
  Active router is local
  Standby router is 192.168.100.2, priority 100 (expires in 8 sec)
  Priority 110 (configured 110)
  Group name is hsrp-V11-10 (default)
Vlan20 - Group 20
  State is Active
    6 state changes, last state change 00:00:37
  Virtual IP address is 192.168.100.126
  Active virtual MAC address is 0000.0C07.AC14
  Local virtual MAC address is 0000.0C07.AC14 (v1 default)
  Hello time 3 sec, hold time 10 sec
  Next hello sent in 2.136 secs
  Preemption enabled
  Active router is local
  Standby router is 192.168.100.66, priority 100 (expires in 8 sec)
  Priority 110 (configured 110)
  Group name is hsrp-V12-20 (default)
Vlan30 - Group 30
  State is Active
    5 state changes, last state change 00:00:19
  Virtual IP address is 192.168.100.190
  Active virtual MAC address is 0000.0C07.AC1E
  Local virtual MAC address is 0000.0C07.AC1E (v1 default)
  Hello time 3 sec, hold time 10 sec
  Next hello sent in 1.111 secs
  Preemption enabled
  Active router is local
  Standby router is 192.168.100.130, priority 100 (expires in 7 sec)
  Priority 110 (configured 110)
  Group name is hsrp-V13-30 (default)
Vlan40 - Group 40
  State is Active
    5 state changes, last state change 00:00:28
  Virtual IP address is 192.168.100.254
  Active virtual MAC address is 0000.0C07.AC28
  Local virtual MAC address is 0000.0C07.AC28 (v1 default)
  Hello time 3 sec, hold time 10 sec
  Next hello sent in 2.759 secs
  Preemption enabled
  Active router is local
  Standby router is 192.168.100.194
  Priority 110 (configured 110)
```

Figure 4.9: HSRP configuration on MLS1 (HQ side)

Meanwhile, MLS2 operates as the Standby as shown in *Figure 4.10*. Preemption is enabled on MLS1 so that it automatically reclaims the Active role after recovery. A virtual IP address is assigned for each VLAN and set as the default gateway for all end devices in that VLAN, ensuring failover is transparent to users.

```
MLS2-HQ>sh
MLS2-HQ2#sh standby
Vlan10 - Group 10
  State is Standby
    6 state changes, last state change 00:00:31
  Virtual IP address is 192.168.100.62
  Active virtual MAC address is 0000.0C07.AC0A
    Local virtual MAC address is 0000.0C07.AC0A (v1 default)
  Hello time 3 sec, hold time 10 sec
    Next hello sent in 0.726 secs
  Preemption disabled
  Active router is 192.168.100.1, priority 110 (expires in 7 sec)
    MAC address is 0000.0C07.AC0A
  Standby router is local
  Priority 100 (default 100)
  Group name is hsrp-Vl1-10 (default)
Vlan20 - Group 20
  State is Standby
    9 state changes, last state change 00:00:48
  Virtual IP address is 192.168.100.126
  Active virtual MAC address is 0000.0C07.AC14
    Local virtual MAC address is 0000.0C07.AC14 (v1 default)
  Hello time 3 sec, hold time 10 sec
    Next hello sent in 2.597 secs
  Preemption disabled
  Active router is 192.168.100.65, priority 110 (expires in 7 sec)
    MAC address is 0000.0C07.AC14
  Standby router is local
  Priority 100 (default 100)
  Group name is hsrp-Vl2-20 (default)
Vlan30 - Group 30
  State is Standby
    6 state changes, last state change 00:00:30
  Virtual IP address is 192.168.100.190
  Active virtual MAC address is 0000.0C07.AC1E
    Local virtual MAC address is 0000.0C07.AC1E (v1 default)
  Hello time 3 sec, hold time 10 sec
    Next hello sent in 0.134 secs
  Preemption disabled
  Active router is 192.168.100.129, priority 110 (expires in 9 sec)
    MAC address is 0000.0C07.AC1E
  Standby router is local
  Priority 100 (default 100)
  Group name is hsrp-Vl3-30 (default)
Vlan40 - Group 40
  State is Standby
    8 state changes, last state change 00:00:47
  Virtual IP address is 192.168.100.254
  Active virtual MAC address is 0000.0C07.AC28
    Local virtual MAC address is 0000.0C07.AC28 (v1 default)
  Hello time 3 sec, hold time 10 sec
    Next hello sent in 0.113 secs
  Preemption disabled
--More--
```

Figure 4.10: HSRP configuration on MLS2 (HQ side)

Following configuration, HSRP states were verified on both switches. MLS1 correctly assumed the Active role and MLS2 assumed the Standby role for all configured VLANs, confirming that the virtual IP addresses are correctly assigned and reachable. Further confirmation is done by typing “show standby brief” to see the summary of the HSRP status as shown in *Figure 4.11*.

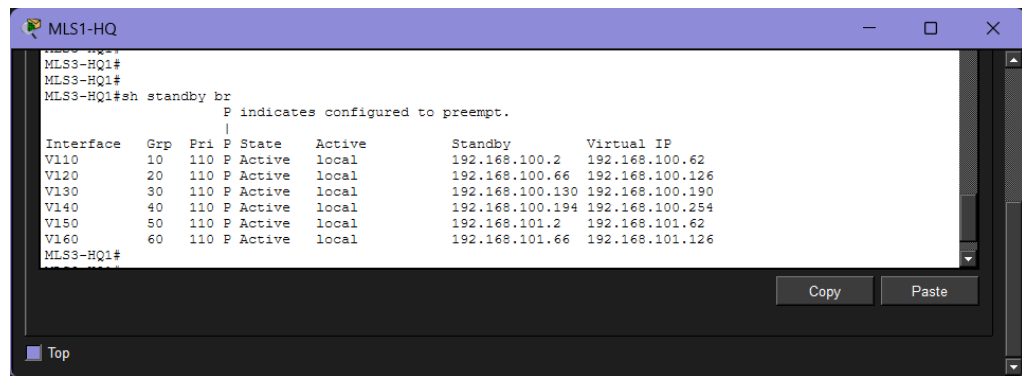


Figure 4.11: HSRP state verification — MLS1 confirmed as Active for all VLANs

4.3.2 RSTP Configuration and Inter-MLS Peer Link

RSTP (IEEE 802.1w) is configured and verified across all Multilayer Switch (MLS) and Access Switch trunk interfaces. A dedicated peer link is established between MLS1 and MLS2 at both the HQ and Branch locations to provide a highly available Layer 2 transit path. Under normal operations, RSTP maintains this core link in a Forwarding state as shown in Figure 4.12.

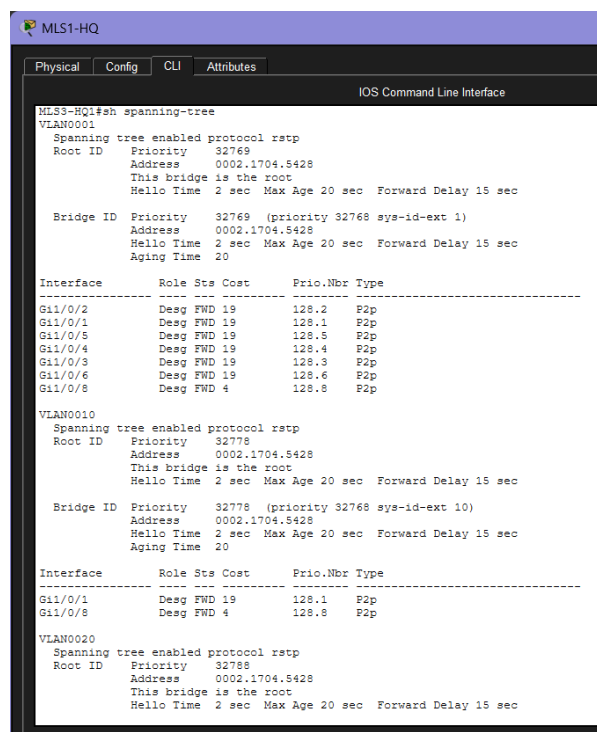


Figure 4.12: Core link (Gi1/0/8) in Forwarding (FWD) state

Meanwhile, loop prevention is efficiently handled by the downstream access switches blocking their redundant alternate uplinks as shown in *Figure 4.13*. In the event of a topology change or physical uplink failure, this peer link provides immediate transit routing, allowing RSTP to converge instantaneously and maintain seamless connectivity without packet loss.

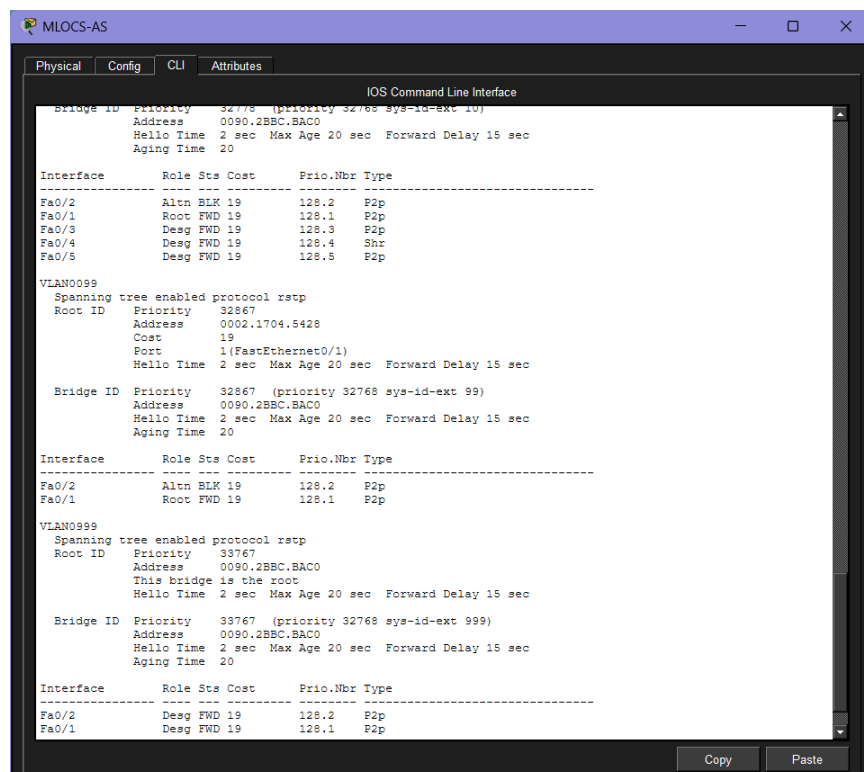


Figure 4.13: Vlan 10 (MLOCS)’s access switch blocking its redundant alternate uplinks.

With HSRP, RSTP and the inter-MLS peer link all verified and operational, the enhanced network is fully configured and ready for performance testing. The results of testing under both normal and failure conditions are presented in Section 4.4.

4.4 Results

This section presents the performance results obtained from testing both the baseline and enhanced networks under controlled failure scenarios. Four failure scenarios are evaluated, each subdivided by whether the MLS–MLS peer link was active or not during the test. The peer link is a dedicated direct link between MLS1 and MLS2 that provides a Layer 2 transit path. The four-performance metrics recorded are Availability, Failover Time, Packet Loss and Packet Delivery Ratio (PDR). Testing was conducted using a continuous ping of 50 packets over a 100-second observation window.

4.4.1 Scenario 1: Link Failure (Without Peer Link)

In this scenario, a link failure is simulated without the MLS–MLS peer link being present. Results show that HSRP + RSTP achieves significantly better performance than HSRP + STP. The RSTP combination reduces failover time from 25 seconds to 7 seconds and cuts packet loss from 4 to 1, yielding a PDR of 98% compared to 92%. The improvement is attributed to RSTP's almost instantaneous convergence, which quickly unblocks an alternate Layer 2 path once the failure is detected, while STP's longer convergence window of up to 30 seconds prolongs the period of traffic disruption. In the absence of any FHRP, the network is entirely unable to recover automatically meaning the default gateway must be reconfigured manually, and all packets are lost until that intervention occurs.

HSRP + STP

Figure 4.14 shows a link is deleted at 00:02:58 mark to simulate link failure. This is the $T_{failure}$.

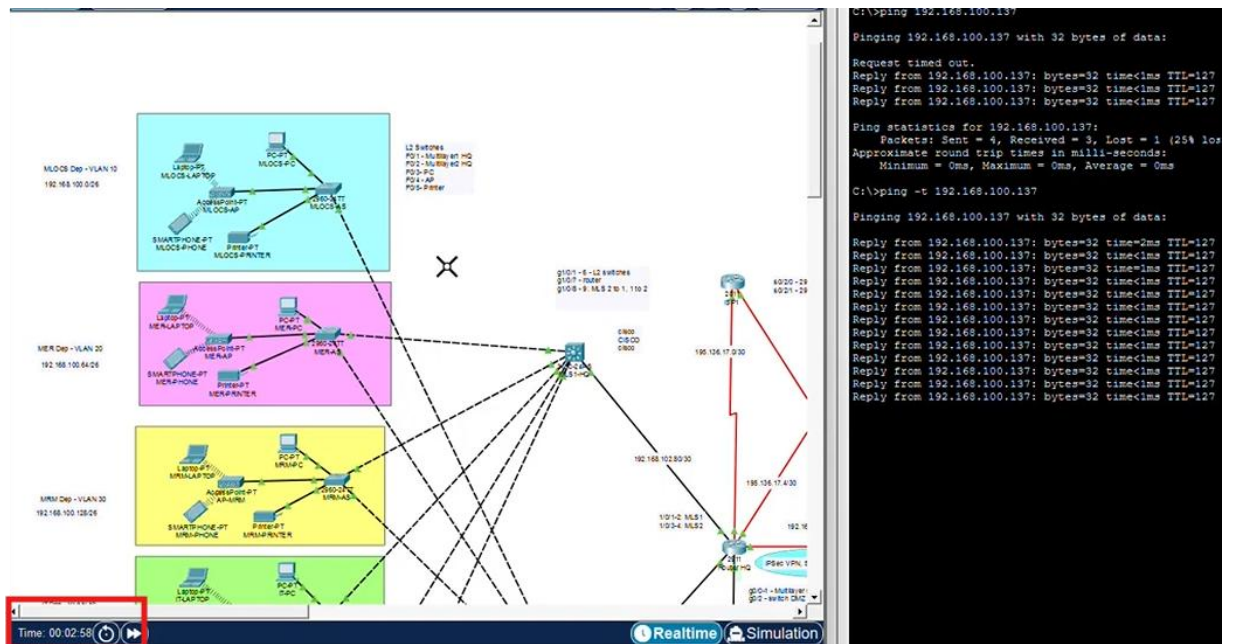


Figure 4.14: HSRP + STP link Failure: Start

Figure 4.15 shows the network recovers at 00:03:23 mark and this is the $T_{ping\ resumes}$. 4 packets were lost during the test.

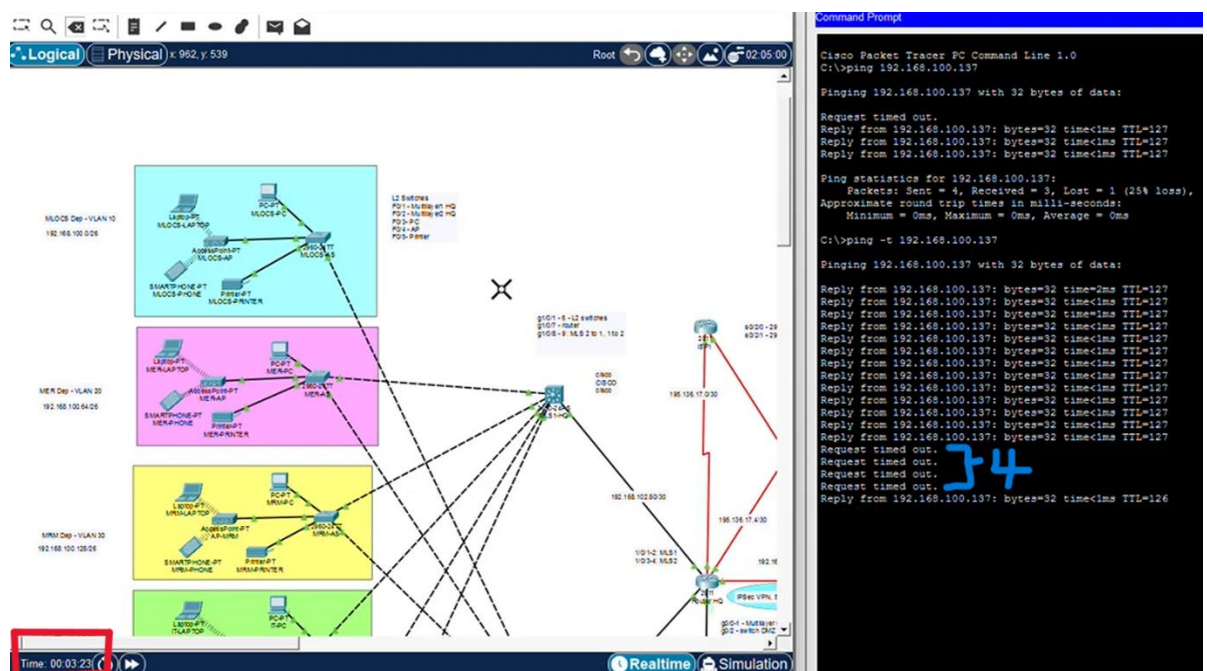


Figure 4.15: HSRP + STP link Failure: End

Observation and Calculation:

$Failover\ Time = 00:03:23 - 00:02:58 = 25\ s$

$Packet\ Loss = 4$

$$Availability = \left(\frac{(100s - 25s)}{100s} \right) \times 100 = 75\%$$

$$PDR = \left(\frac{(50 - 4)}{50} \right) \times 100 = 92\%$$

HSRP + RSTP

Figure 4.16 shows a link is deleted at 00:11:55 mark to simulate link failure. This is the

$T_{failure}$.

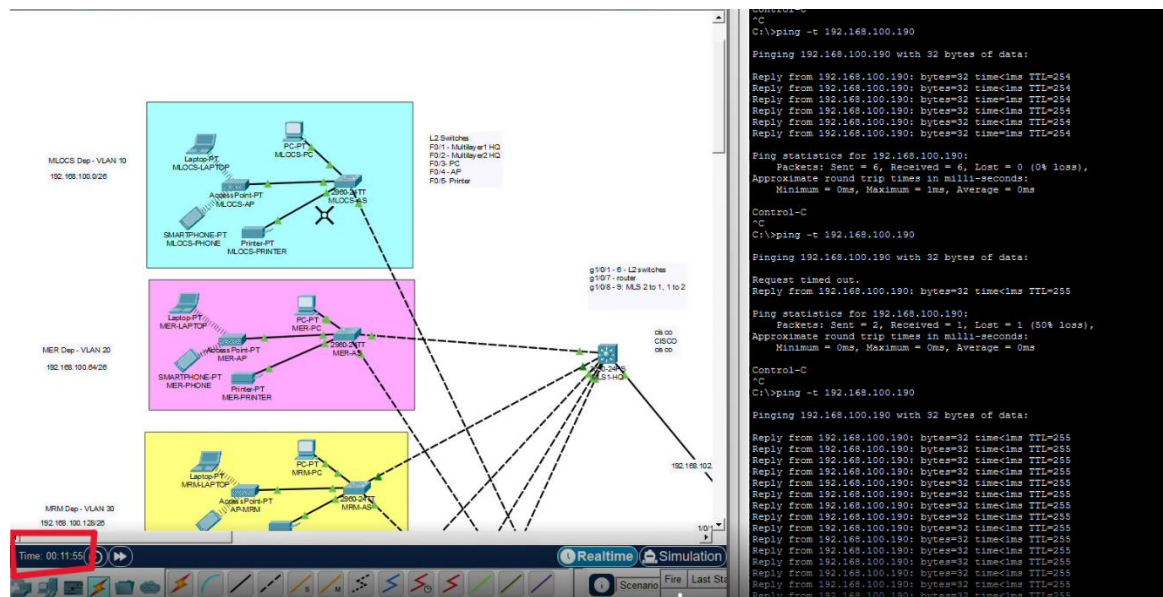


Figure 4.16: HSRP + RSTP link Failure: Start

Figure 4.17 shows the network recovers at 00:12:02 mark and this is the $T_{ping\ resumes}$. 1 packet was lost during the test.

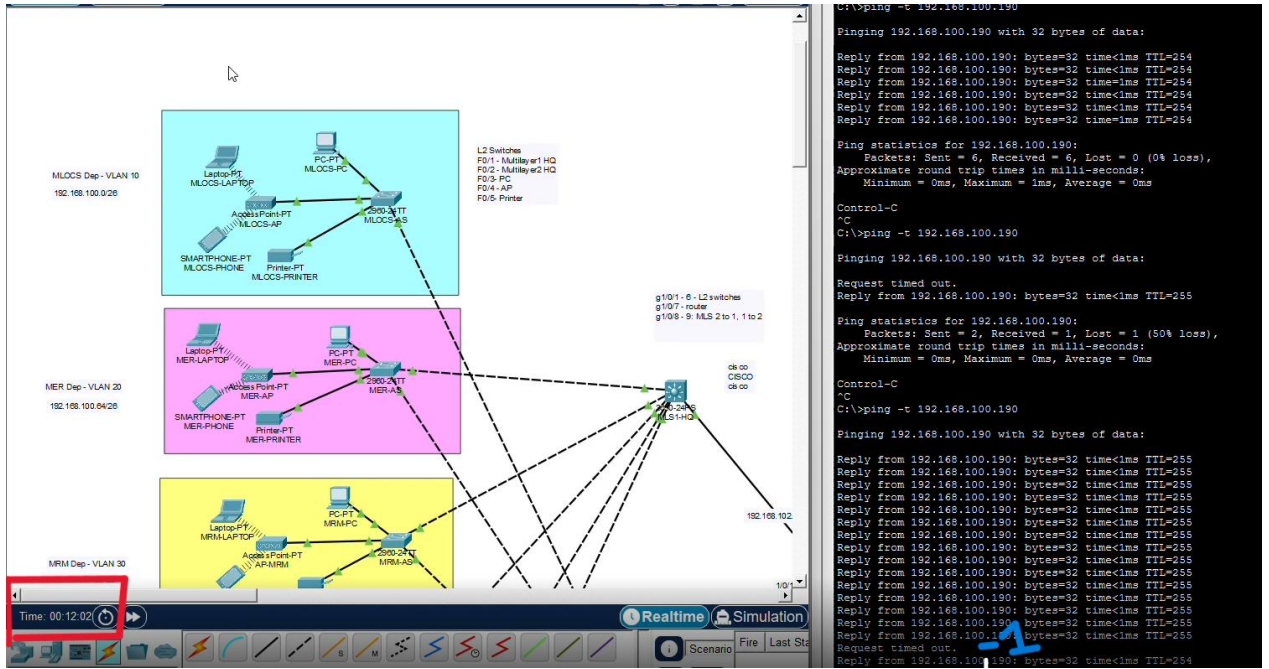


Figure 4.17: HSRP + RSTP link Failure: End

Observation and Calculation:

$$\text{Failover Time} = 00:12:02 - 00:11:55 = 7 \text{ s}$$

$$\text{Packet Loss} = 4$$

$$\text{Availability} = \left(\frac{(100\text{s} - 7\text{s})}{100\text{s}} \right) \times 100 = 93\%$$

$$\text{PDR} = \left(\frac{(50 - 1)}{50} \right) \times 100 = 98\%$$

Protocol Combination	Availability (%)	Failover Time (s)	Packet Loss (Sent = 50)	PDR (%)
HSRP + STP	75%	25 s	4	92%
HSRP + RSTP	93%	7 s	1	98%
No HSRP	N/A — manual intervention required			

Table 4.3: Performance Metrics: Link Failure (Without Peer Link)

4.4.2 Scenario 2: Link Failure (With Peer Link)

This scenario introduces the MLS–MLS peer link into the same link-failure condition. The peer link provides an immediate Layer 2 transit path between the two distribution-layer switches, allowing RSTP to leverage it the instant the primary uplink fails. With HSRP + RSTP, availability reaches approximately 99%, failover time drops below 1 second, and zero packet loss is recorded, which means it achieves a PDR of 100%. This is the best result across all tested scenarios and demonstrates that the combination of a peer link with RSTP produces near-seamless recovery that is effectively invisible to end users.

In contrast, HSRP + STP with the peer link performs worse than in Scenario 1. The additional peer link increases Layer 2 topology complexity, forcing STP to re-evaluate more port transitions before converging. This extended STP reconvergence period raises failover time to 31 seconds, increases packet loss to 5, and reduces availability to 69%. This result underscores that without RSTP, introducing a peer link can degrade performance rather than improving it.

HSRP + STP

Figure 4.18 shows a link is deleted at 00:43:14 mark to simulate link failure. This is the $T_{failure}$.

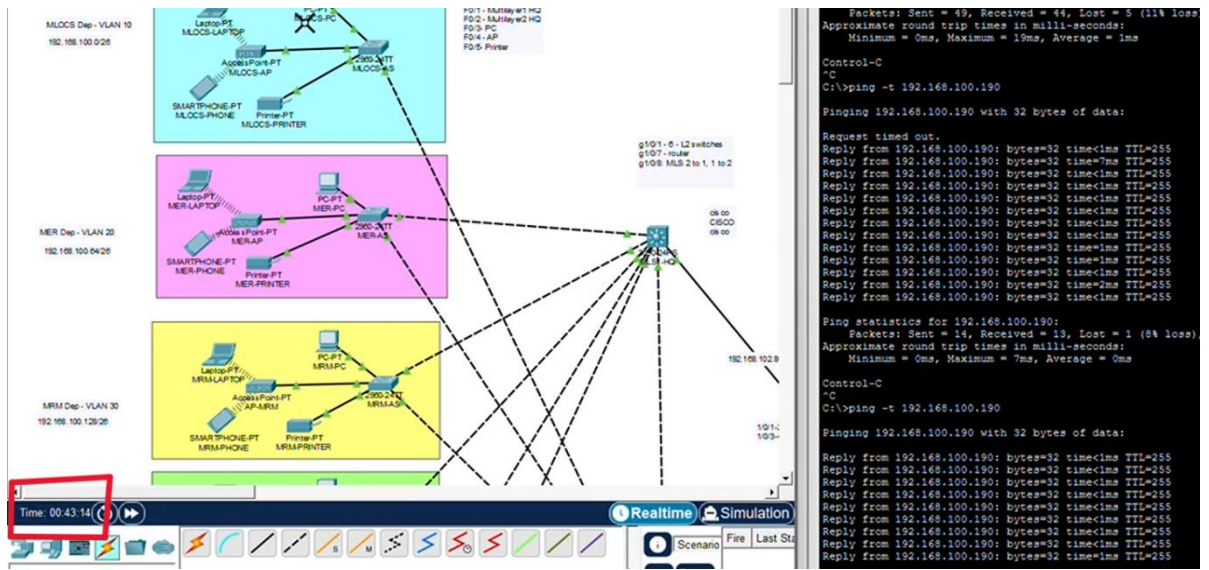


Figure 4.18: HSRP + STP + Peer Link, Link Failure: Start

Figure 4.19 shows the network recovers at 00:43:45 mark and this is the $T_{ping\ resumes}$. 5 packets were lost during the test.

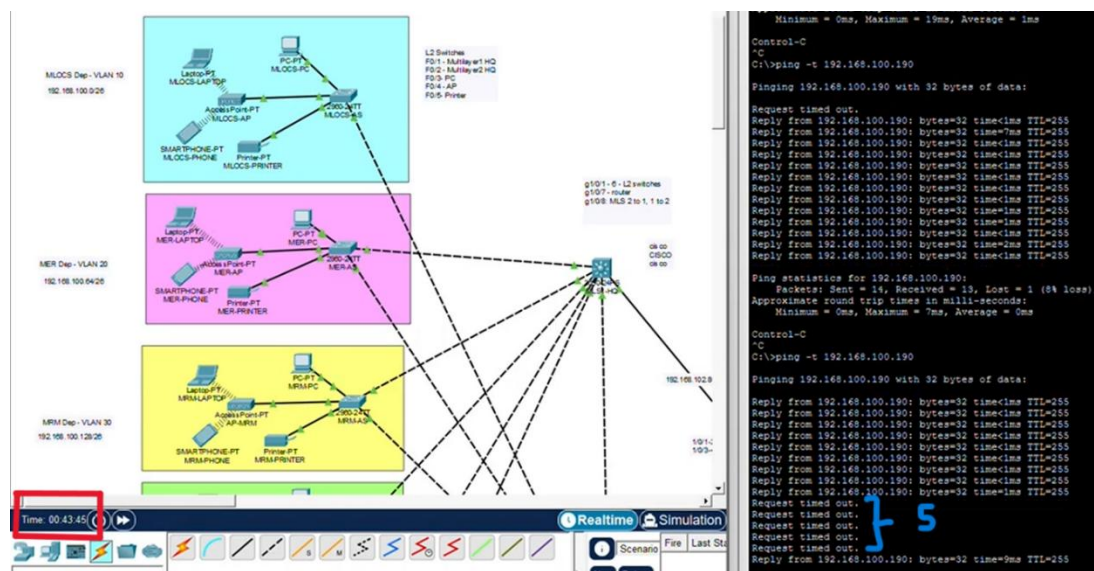


Figure 4.19: HSRP + STP + Peer Link, Link Failure: End

Observation and Calculation:

$$Failover\ Time = 00:43:45 - 00:43:14 = 31\ s$$

$$Packet\ Loss = 5$$

$$Availability = \left(\frac{(100s - 31s)}{100s} \right) \times 100 = 69\%$$

$$PDR = \left(\frac{(50 - 5)}{50} \right) \times 100 = 90\%$$

HSRP + RSTP

Figure 4.20 shows a link is deleted at 00:11:56 mark to simulate link failure. This is the

$T_{failure}$.

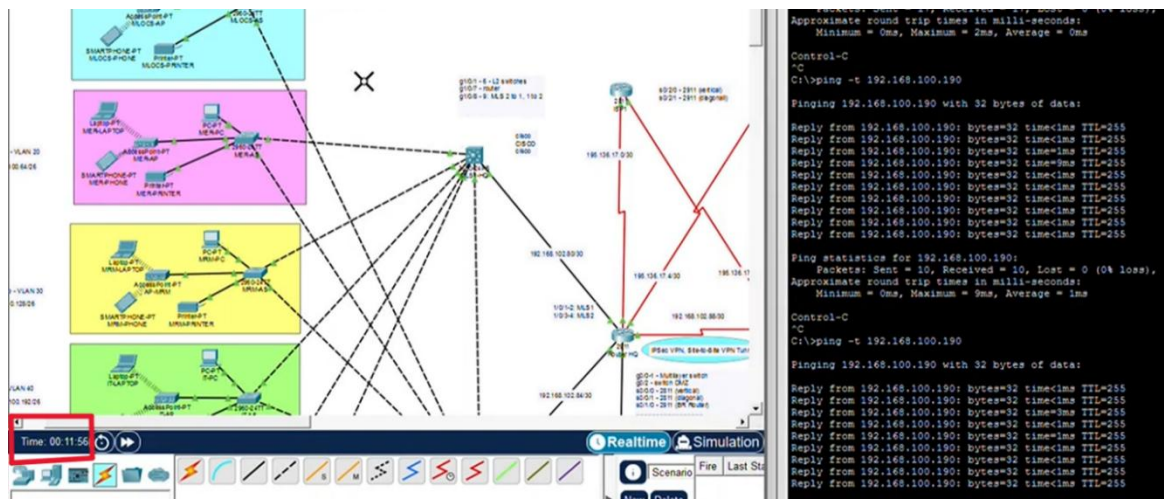


Figure 4.20: HSRP + RSTP + Peer Link, Link Failure: Start

Figure 4.21 shows the network shows no sign of going down that even after reaching

00:12:05 mark, no “Request timed out” message appeared. $T_{ping\ resumes} = 00:11:57$,

Packet Loss = 0.

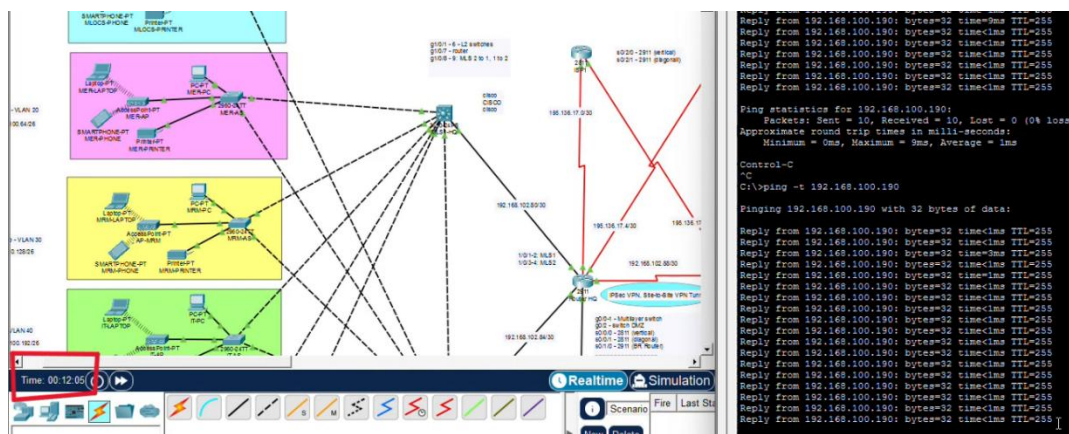


Figure 4.21: HSRP + RSTP + Peer Link, Link Failure: End

Observation and Calculation:

$$\text{Failover Time} = 00:11:57 - 00:11:56 = 1s$$

$$\text{Packet Loss} = 0$$

$$\text{Availability} = \left(\frac{(100s - 0s)}{100s} \right) \times 100 = 100\% = \sim 99\%$$

$$\text{PDR} = \left(\frac{(50 - 0)}{50} \right) \times 100 = 100\%$$

Protocol Combination	Availability (%)	Failover Time (s)	Packet Loss (Sent = 50)	PDR (%)
HSRP + STP	69%	31 s	5	90%
HSRP + RSTP	~99%	< 1 s	0	100%

Table 4.4: Performance Metrics: Link Failure (With Peer Link)

4.4.3 Scenario 3: Active MLS Failure (Without Peer Link)

In this scenario, the active MLS (MLS1) is brought down entirely to simulate a device failure, without the peer link present. Both HSRP + STP and HSRP + RSTP record identical results: **88% availability, a 12-second failover time, 2 packets lost and a PDR of 96%**. This outcome is expected because the failover in this scenario is governed entirely by HSRP's hold timer causing the standby MLS2 to wait for three consecutive missed hello messages before promoting itself to the Active role. Since both configurations use the same HSRP timers, the gateway switchover duration is identical regardless of which Layer 2 spanning tree protocol is in use. Layer 2 convergence is not the bottleneck here; HSRP's detection mechanism is.

HSRP + STP

Figure 4.22 shows the HSRP active MLS (MLS1) power source is removed at 00:10:33 mark to simulate MLS failure. This is the $T_{failure}$.

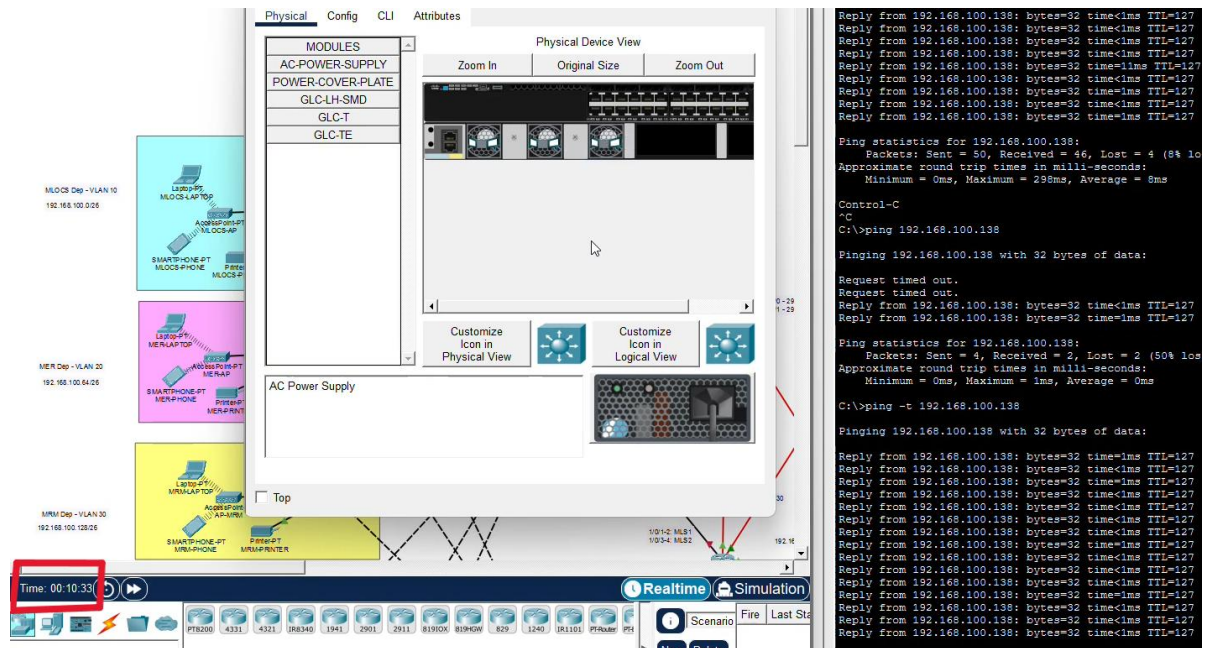


Figure 4.22: HSRP + STP, MLS Failure: Start

Figure 4.23 shows the network recovers at 00:10:45 mark and this is the $T_{ping\ resumes}$. 2 packets were lost during the test.

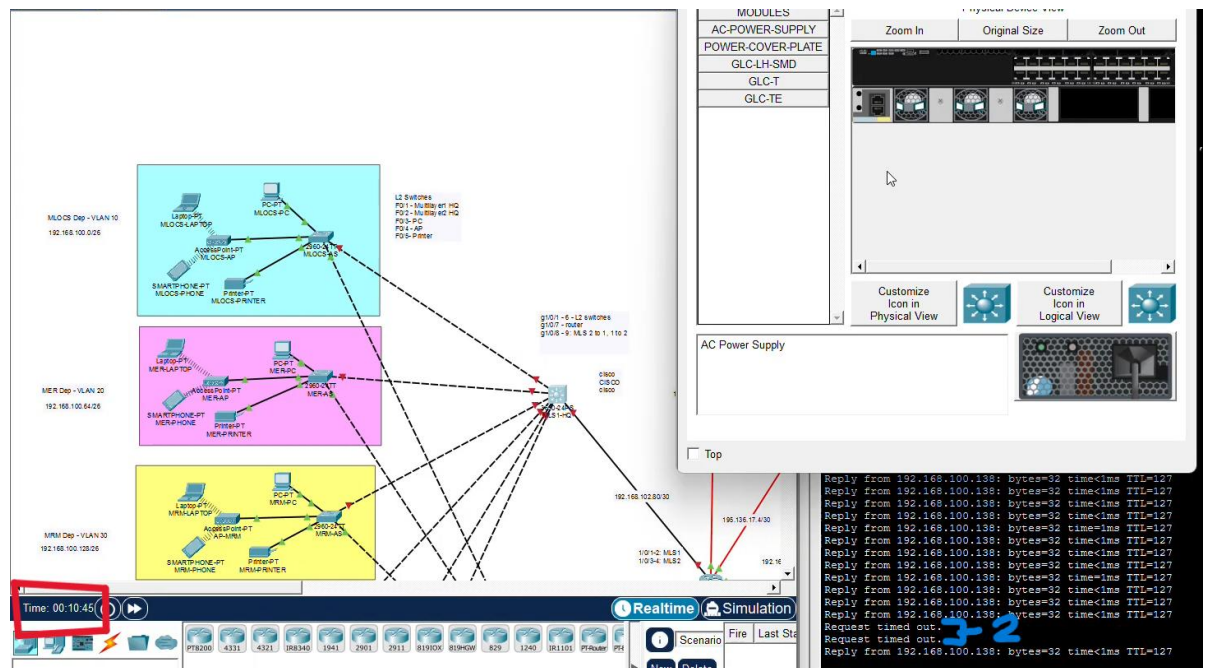


Figure 4.23: HSRP + STP, MLS Failure: End

Observation and Calculation:

$$Failover\ Time = 00:10:45 - 00:10:33 = 12\ s$$

$$Packet\ Loss = 2$$

$$Availability = \left(\frac{(100s - 12s)}{100s} \right) \times 100 = 88\%$$

$$PDR = \left(\frac{(50 - 2)}{50} \right) \times 100 = 96\%$$

HSRP + RSTP

Figure 4.24 shows the HSRP active MLS (MLS1) power source is removed at 00:25:28 mark to simulate MLS failure. This is the $T_{failure}$.

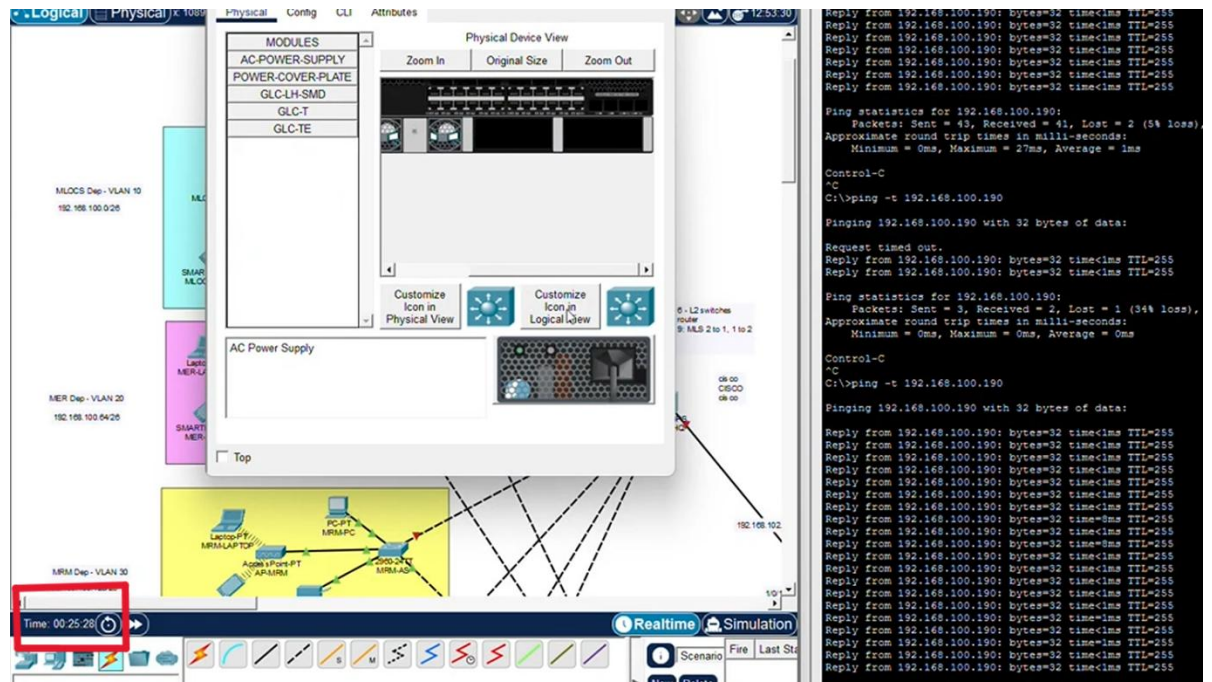


Figure 4.24: HSRP + RSTP, MLS Failure: Start

Figure 4.25 shows the network recovers at 00:25:40 mark and this is the $T_{ping\ resumes}$. 2 packets were lost during the test.

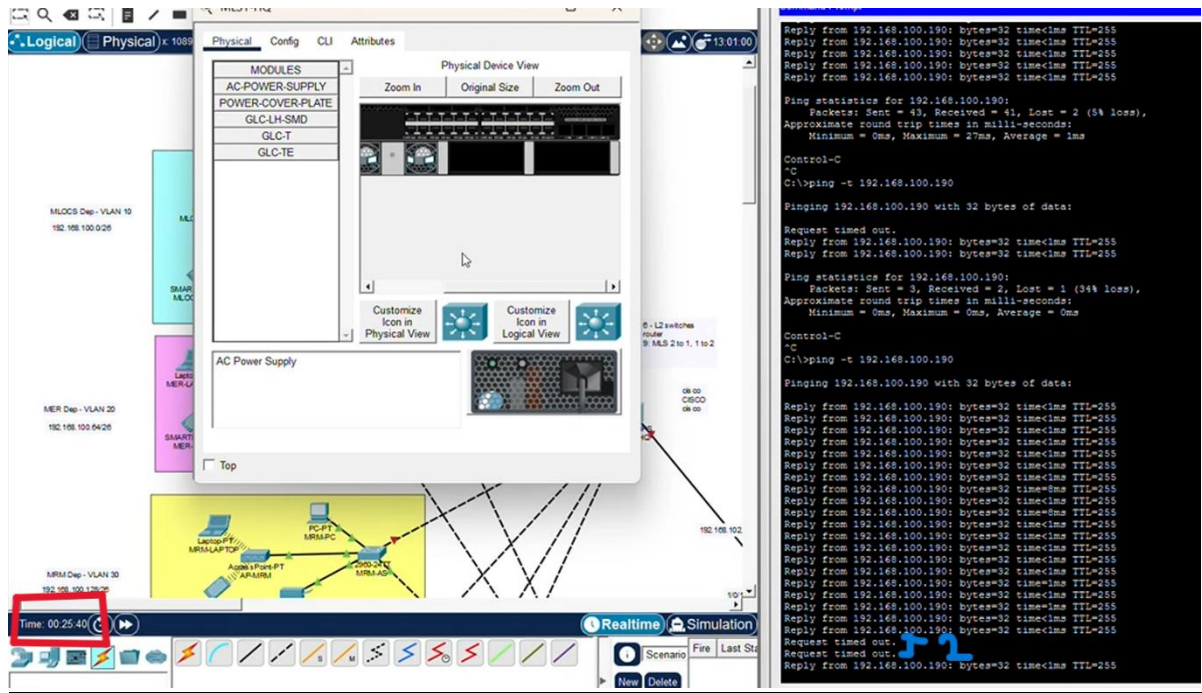


Figure 4.25: HSRP + RSTP, MLS Failure: End

Observation and Calculation:

$$\text{Failover Time} = 00:25:40 - 00:25:28 = 12 \text{ s}$$

$$\text{Packet Loss} = 2$$

$$\text{Availability} = \left(\frac{(100\text{s} - 12\text{s})}{100\text{s}} \right) \times 100 = 88\%$$

$$\text{PDR} = \left(\frac{(50 - 2)}{50} \right) \times 100 = 96\%$$

Protocol Combination	Availability (%)	Failover Time (s)	Packet Loss (Sent = 50)	PDR (%)
HSRP + STP	88%	12 s	2	96%
HSRP + RSTP	88%	12 s	2	96%

Table 4.5: Performance Metrics: Active MLS Failure (Without Peer Link)

4.4.4 Scenario 4: Active MLS Failure (With Peer Link)

This scenario repeats the active MLS failure with the peer link present. HSRP + RSTP

maintains the same result as Scenario 3 which is **88% availability, 12-second failover, 2 packets lost and 96% PDR** since the failover is again dominated by the HSRP hold timer and RSTP's role is minimal once the peer link is active. However, HSRP + STP degrades considerably compared to Scenario 3. With the peer link in place, STP must re-elect port roles and recalculate path costs for the more complex topology after MLS1 goes down. This reconvergence is substantially slower than in Scenario 3, driving **failover time up to 30 seconds, packet loss to 5 and availability down to 70%.**

This finding highlights a key distinction: the peer link benefits RSTP-based configurations significantly during link failures (Scenario 2) but introduces additional STP reconvergence overhead during device failures when combined with STP. RSTP's ability to transition ports to Forwarding state without waiting through Listening and Learning phases means it remains unaffected by this added complexity.

HSRP + STP

Figure 4.26 shows the HSRP active MLS (MLS1) power source is removed at 00:51:45 mark to simulate MLS failure. This is the $T_{failure}$.

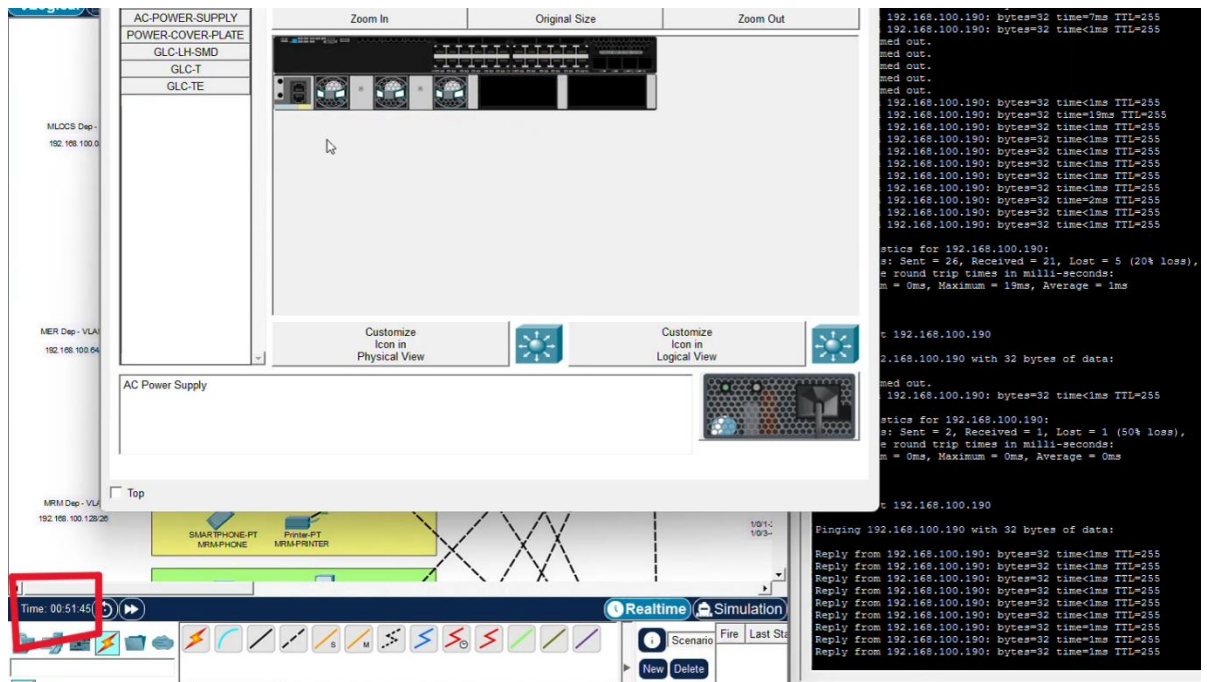


Figure 4.26: HSRP + STP + Peer Link, MLS Failure: Start

Figure 4.27 shows the network recovers at 00:52:15 mark and this is the $T_{ping\ resumes} \cdot 5$ packets were lost during the test.

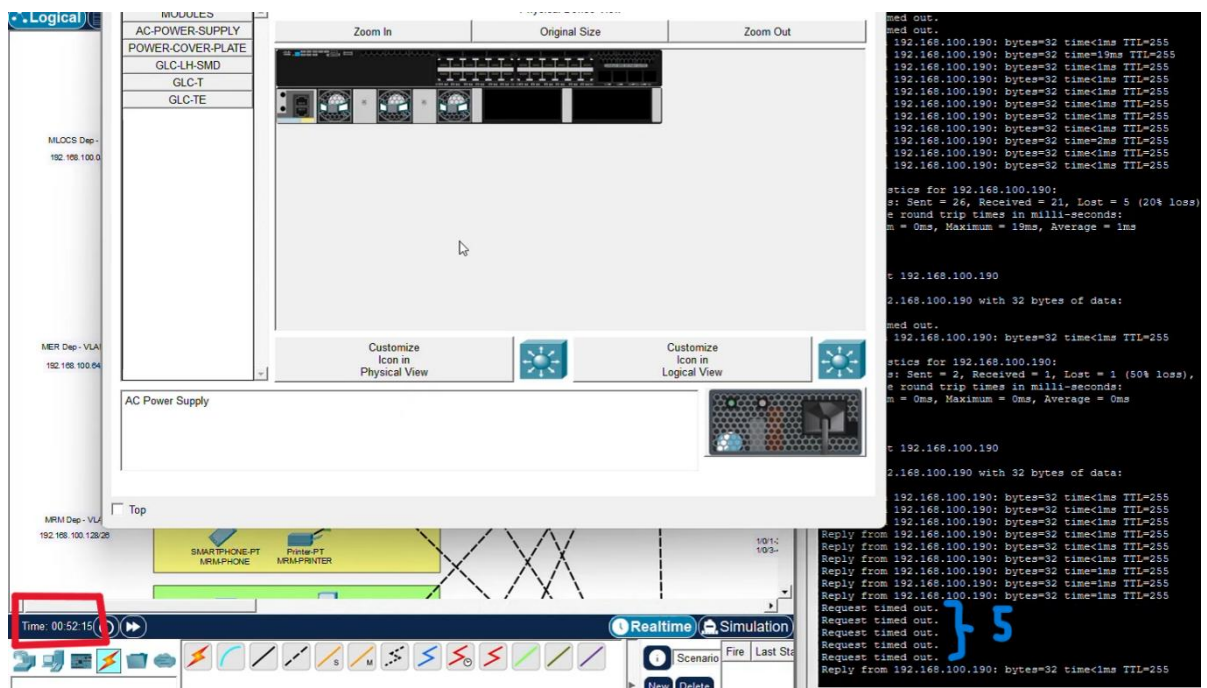


Figure 4.27: HSRP + STP + Peer Link, MLS Failure: End

Observation and Calculation:

$$\text{Failover Time} = 00:52:15 - 00:51:45 = 30 \text{ s}$$

$$\text{Packet Loss} = 5$$

$$\text{Availability} = \left(\frac{(100\text{s} - 30\text{s})}{100\text{s}} \right) \times 100 = 70\%$$

$$\text{PDR} = \left(\frac{(50 - 5)}{50} \right) \times 100 = 90\%$$

HSRP + RSTP

Figure 4.28 shows the HSRP active MLS (MLS1) power source is removed at 00:22:16 mark to simulate MLS failure. This is the T_{failure} .

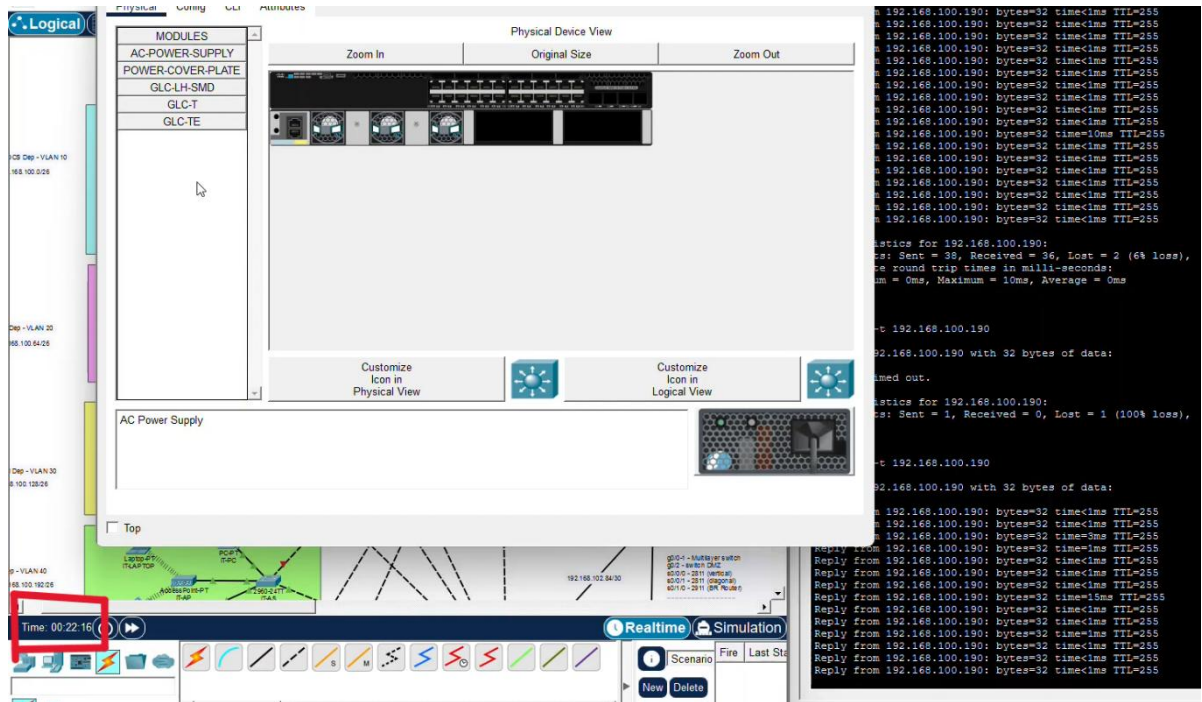


Figure 4.28: HSRP + RSTP + Peer Link, MLS Failure: Start

Figure 4.29 shows the network recovers at 00:22:28 mark and this is the $T_{\text{ping resumes}}$. 2 packets were lost during the test.

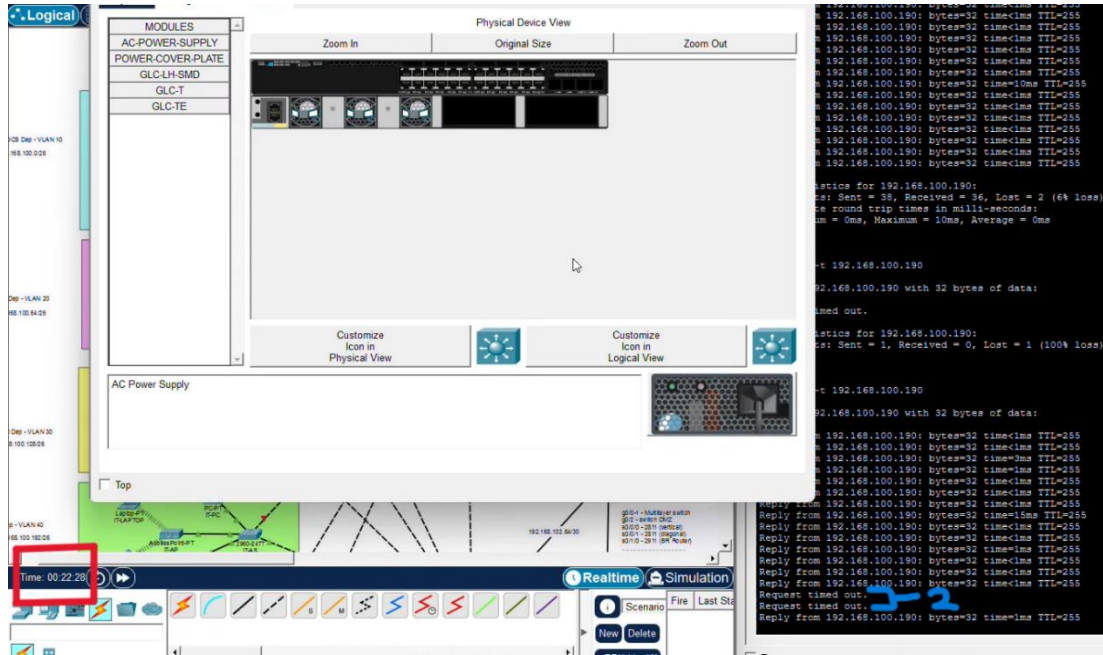


Figure 4.29: HSRP + RSTP + Peer Link, MLS Failure: End

Observation and Calculation:

$$\text{Failover Time} = 00:22:28 - 00:22:16 = 12 \text{ s}$$

$$\text{Packet Loss} = 2$$

$$\text{Availability} = \left(\frac{(100\text{s} - 12\text{s})}{100\text{s}} \right) \times 100 = 88\%$$

$$\text{PDR} = \left(\frac{(50 - 2)}{50} \right) \times 100 = 96\%$$

Protocol Combination	Availability (%)	Failover Time (s)	Packet Loss (Sent = 50)	PDR (%)
HSRP + STP	70%	30 s	5	90%
HSRP + RSTP	88%	12 s	2	96%

Table 4.6: Performance Metrics: Active MLS Failure (With Peer Link)

4.4.5 Summary of Results

The following table and graphs consolidate all performance metrics across all four scenarios and both protocol combinations, enabling a direct cross-scenario comparison.

From Table 4.7, we can conclude, across all scenarios, HSRP + RSTP consistently outperforms or matches HSRP + STP. The most significant gains are observed in Scenario 2, where the peer link and RSTP together produce near-zero failover with 100% PDR. When failover is governed by HSRP timer expiry rather than Layer 2 convergence (Scenarios 3 and 4 without peer link), both configurations perform identically, confirming that HSRP's hold timer is the limiting factor in device-level failures. The peer link amplifies the performance gap between the two protocol configurations: it dramatically improves HSRP + RSTP performance during link failures but introduces reconvergence penalties for HSRP + STP during both link and device failures, making the choice of spanning tree protocol a critical design decision when a peer link is deployed.

Failure Scenario	Protocol Config	Availability (%)	Failover Time	Packet Loss	PDR (%)
Link Failure (No Peer Link)	HSRP + STP	75%	25 s	4 / 50	92%
Link Failure (No Peer Link)	HSRP + RSTP	93%	7 s	1 / 50	98%
Link Failure (With Peer Link)	HSRP + STP	69%	31 s	5 / 50	90%
Link Failure (With Peer Link)	HSRP + RSTP	~99%	< 1 s	0 / 50	100%
Active MLS Failure (No Peer Link)	HSRP + STP	88%	12 s	2 / 50	96%
Active MLS Failure (No Peer Link)	HSRP + RSTP	88%	12 s	2 / 50	96%
Active MLS Failure (With Peer Link)	HSRP + STP	70%	30 s	5 / 50	90%
Active MLS Failure (With Peer Link)	HSRP + RSTP	88%	12 s	2 / 50	96%

Table 4.7: Summary of All Performance Metrics Across All Failure Scenarios

Figure 4.30 shows that the HSRP + RSTP configuration achieved higher network availability than HSRP + STP in most failure scenarios, with the highest availability (~99%) recorded during link failure with a peer link.

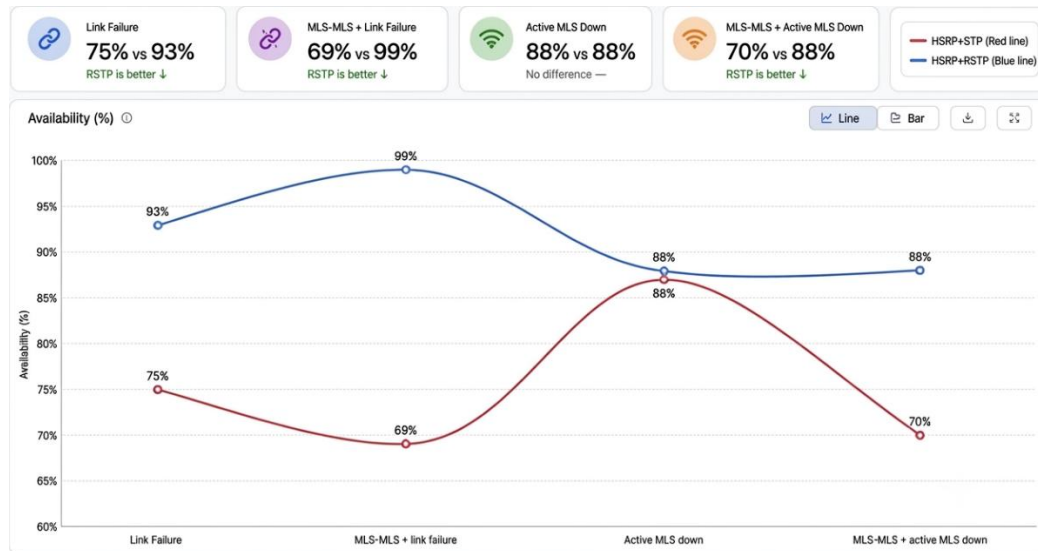


Figure 4.30: Availability Graph

Figure 4.31 shows that HSRP + RSTP consistently provided faster failover times than HSRP + STP, with the shortest recovery time being less than 1 second when a peer link was used during a link failure.

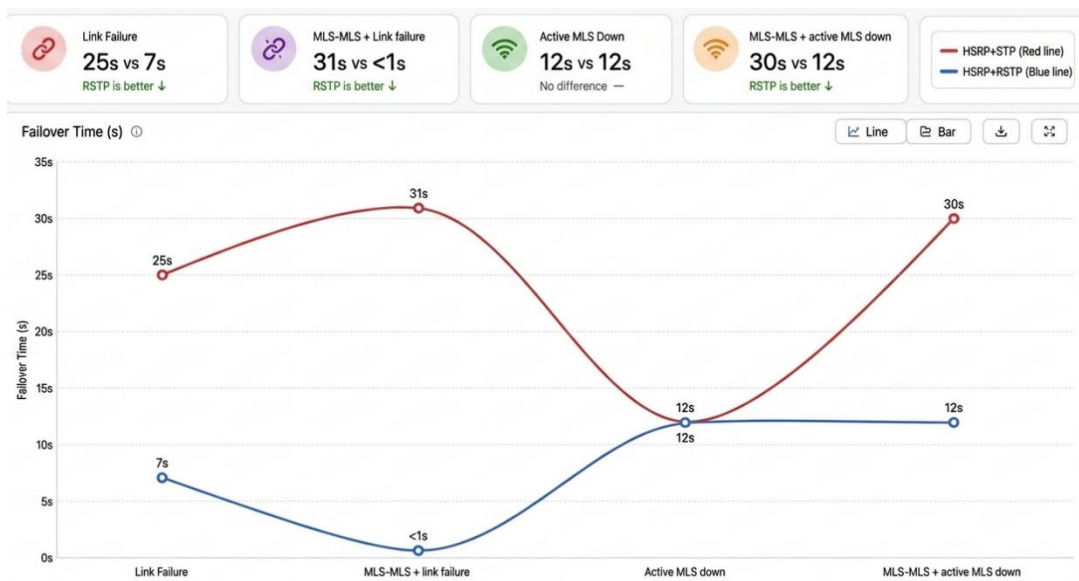


Figure 4.31: Failover Time Graph

Figure 4.32 shows that HSRP + RSTP resulted in lower packet loss across all scenarios, achieving zero packet loss during the link failure scenario with a peer link.



Figure 4.32: Packet Loss Graph

Figure 4.33 shows that HSRP + RSTP maintained a higher Packet Delivery Ratio (PDR) than HSRP + STP, reaching 100% PDR when the peer link and RSTP were combined.

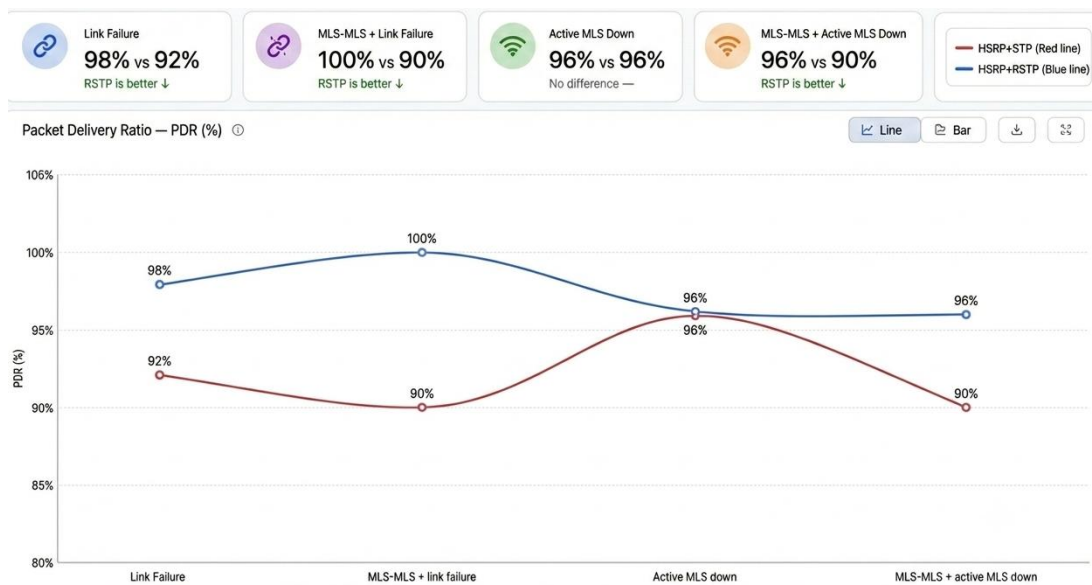


Figure 4.33: PDR Graph

4.5 Summary of Chapter

This chapter documented the full implementation and evaluation of the proposed hospital network enhancement. The baseline network, replicating the original Medicine City Hospital design by Jassim (2024), was first established and verified to be fully functional before any failure testing was conducted. The enhanced network was then configured by adding HSRP for default gateway redundancy and RSTP with a peer link between MLS1 and MLS2 for Layer 2 resilience, without altering any existing baseline protocols.

Performance testing was conducted across four failure scenarios: link failure and active MLS failure, each tested with and without the MLS–MLS peer link. Results demonstrated that HSRP + RSTP consistently achieves superior or equivalent performance compared to HSRP + STP across all scenarios. The most critical finding is that the combination of the peer link with RSTP in Scenario 2 produced approximately 99% availability, sub-second failover, zero packet loss and a PDR of 100%, representing the closest outcome to seamless, transparent failover. The peer link's role was clarified as a Layer 2 transit path and its interaction with the spanning tree protocol was shown to be a decisive factor in determining failover performance. These results collectively validate the effectiveness of the proposed redundancy mechanisms in enhancing hospital network resilience.

CHAPTER 5

CONCLUSION

5.1 Introduction

This chapter presents the conclusion of the project, summarising how the objectives were achieved, the contributions of the work, and recommendations for future research and real-world deployment. The chapter reflects on findings from the implementation and evaluation phase and contextualises them within the broader goal of improving hospital network resilience through redundancy protocols.

5.2 Objective Achievement

This project set out **three objectives**, all of which have been addressed through the design, implementation and evaluation of the enhanced hospital network.

The **first objective** was to study the role of redundant protocol techniques in improving network resilience and availability within a hospital environment. This was accomplished through an extensive literature review spanning thirteen papers covering HSRP, RSTP, STP, EtherChannel and LACP in various enterprise and critical infrastructure contexts. The review established that HSRP is the preferred First Hop Redundancy Protocol for complex VLAN environments due to its fastest average time consumption and highest packet transfer rate among FHRP options, while RSTP's near-instantaneous convergence makes it substantially superior to traditional STP for Layer 2 redundancy in time-sensitive networks.

The **second objective** was to develop a resilient hospital network design using HSRP and RSTP implemented in Cisco Packet Tracer. The baseline network, faithfully

replicating the Medicine City Hospital design by Jassim (2024), was augmented with HSRP configured per VLAN on MLS1 and MLS2, with MLS1 as the Active switch and pre-emption enabled, and RSTP enabled across all trunk interfaces alongside a dedicated peer link between MLS1 and MLS2. All original protocols (OSPF, DHCP, SSH, ACL, NAT, PAT and IPsec Site-to-Site VPN) were preserved without modification, demonstrating that the redundancy layer can be introduced non-disruptively over an existing production-like configuration.

The **third objective** was to compare the enhanced network performance against the existing network design using availability, failover time, packet loss and PDR. Four failure scenarios were tested across two protocol configurations. Results confirmed that HSRP + RSTP outperforms HSRP + STP in every scenario where Layer 2 convergence is a factor and is never worse. The most compelling result came from Scenario 2, where HSRP + RSTP with the peer link achieved approximately 99% availability, sub-second failover, zero packet loss and a PDR of 100%, demonstrating that the combination of HSRP, RSTP and a peer link can eliminate perceptible service disruption during link failures.

5.3 Key Findings

Several important findings emerged from this project that extend beyond the specific numbers recorded in the results section.

RSTP is essential whenever a peer link is present. The results showed that introducing the peer link improved HSRP + RSTP performance dramatically but actively degraded HSRP + STP performance in both link-failure and device-failure scenarios. STP's slower convergence algorithm cannot efficiently process the additional topology complexity

introduced by the peer link, whereas RSTP's proposal-agreement mechanism handles it seamlessly. This means that deploying a peer link without upgrading to RSTP is counterproductive and may worsen network availability compared to a simpler single-uplink configuration.

HSRP timer expiry is the primary bottleneck during device-level failures. In all active MLS failure scenarios, both protocol combinations recorded a 12-second failover time, determined entirely by the HSRP default hold timer of 10 seconds plus propagation delay. This confirms that reducing failover time during device failures requires tuning HSRP hello and hold timers rather than optimising the spanning tree configuration. Tuning these timers would be a high-impact, low-risk enhancement for future work.

The absence of any FHRP renders the network unrecoverable without manual intervention. Without HSRP, a link or device failure that takes down the active gateway causes all traffic to cease until an administrator manually reassigns the default gateway on affected devices. In a hospital context, this would translate directly into loss of access to Electronic Health Records and critical clinical systems, potentially endangering patient safety. This confirms the non-negotiable necessity of FHRP in any healthcare network design.

5.4 Contributions

This project makes two primary contributions. First, it provides a practical validation of HSRP and RSTP redundancy mechanisms in a hospital-specific network context, demonstrating measurable improvements in availability, failover time, packet loss and PDR relative to a non-redundant baseline. Second, it isolates and quantifies the interaction effect between the MLS–MLS peer link and the choice of spanning tree

protocol, which is a relationship that is not always explicitly addressed in existing literature and shows that this interaction is a critical design variable that determines whether the peer link enhances or degrades performance.

5.5 Limitations

Several limitations of this work should be acknowledged when interpreting and applying the results. The simulation was conducted entirely in Cisco Packet Tracer 9.0.0, which approximates but does not fully replicate real-world network behaviour. Physical hardware introduces additional factors such as hardware queuing delays, cable propagation characteristics and asynchronous clock drift that may affect failover timing. The results should therefore be treated as indicative rather than absolute benchmarks for production deployment.

The observation window for each test was fixed at 100 seconds with a 50-packet ping, which may not adequately capture transient effects or variability across repeated trials. A more rigorous evaluation would involve multiple repetitions of each scenario, statistical analysis of variance, and longer observation periods to detect anomalous behaviours. Furthermore, HSRP hello and hold timers were left at their default values of 3 seconds and 10 seconds respectively throughout all tests. Reduced timers would lower failover time in device-failure scenarios but may introduce instability in a real environment, a trade-off that was outside the scope of this project.

The security components of the network: ACL, NAT, PAT, SSH and IPSec VPN, were verified as functional but were not subjected to failure analysis. Advanced enterprise security mechanisms such as stateful firewalls were also excluded, as the project focused on availability and redundancy rather than security architecture.

5.6 Recommendations for Future Work

Several directions are recommended to extend and strengthen this work. Tuning HSRP hello and hold timers to sub-second values (e.g. hello 200ms, hold 600ms) would significantly reduce device-failure failover time beyond the 12 seconds observed and bring availability closer to the Five Nines standard targeted by critical healthcare infrastructure. This should be evaluated against the risk of false failovers caused by network jitter.

Implementing EtherChannel with LACP on the uplinks between access switches and distribution-layer MLS devices would add link-level redundancy and increase available bandwidth, complementing the gateway and Layer 2 redundancy already provided by HSRP and RSTP. This combination was highlighted in multiple reviewed papers as an effective integrated solution for enterprise-grade resilience.

Future work should also explore replacing the Collapsed Core (two-tier) architecture with a full three-tier design incorporating a dedicated Core layer, which would provide additional path diversity and support greater network scale. In a real hospital deployment, this would also create natural segmentation between clinical and administrative traffic, improving both availability and security posture.

Finally, hardware-based validation using physical Cisco switches and routers would provide a more accurate characterisation of real-world failover behaviour. Pairing this with live application traffic such as simulated EHR queries or medical imaging transfers rather than ICMP ping tests would give a more clinically meaningful measure of network performance under failure conditions.

5.7 Conclusion

This project successfully demonstrated that implementing HSRP and RSTP redundancy protocols in a hospital network significantly enhances its resilience and availability. The enhanced network, built on an existing Medicine City Hospital design, consistently outperformed the baseline across all tested failure scenarios, with the most outstanding result being near-perfect availability and zero packet loss when HSRP and RSTP were combined with a peer link during a link-failure event. The findings confirm that RSTP is not merely an incremental improvement over STP but a fundamentally different protocol whose rapid convergence is essential for any network topology that includes redundant inter-switch paths. They also establish that HSRP is a necessary first layer of defence against gateway-level failures in critical healthcare environments. Together, these protocols form a complementary and cost-effective redundancy stack that can be applied to existing hospital network infrastructure without disrupting current services, offering a practical pathway toward the high-availability standards that patient safety demands.

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